

Quantum machine learning for complex systems: paradigms, applications, and challenges

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Abstract

Quantum machine learning (QML) is rapidly transitioning from theoretical promise to practical relevance across data-intensive scientific domains. In this review, we provide a structured overview of recent advances that bridge foundational quantum learning principles with real-world applications. We survey foundational QML paradigms, including variational quantum algorithms, quantum kernel methods, and neural-network quantum states, with emphasis on their applicability to complex quantum systems. We examine neural-network quantum states as expressive variational models for correlated matter, non-equilibrium dynamics, and open quantum systems, and discuss fundamental challenges associated with training and sampling. Recent advances in quantum-enhanced sampling and diagnostics of learning dynamics, including information-theoretic tools, are reviewed as mechanisms for improving scalability and trainability. The review further highlights application-driven QML frameworks in drug discovery, cancer biology, and agro-climate modeling, where data complexity and constraints motivate hybrid quantum–classical approaches. We conclude with a discussion of federated quantum machine learning as a route to distributed, privacy-preserving quantum learning. Overall, this review presents a unified perspective on the opportunities and limitations of QML for complex systems.

Keywords: *quantum machine learning, variational quantum algorithms, neural-network quantum states, federated quantum learning*

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1. Introduction

Complex systems arise across a wide range of scientific domains, including condensed-matter and chemical systems, biological and social networks, and climate and agro-environmental dynamics, where macroscopic behavior emerges from the collective interaction of many microscopic degrees of freedom. Such systems are typically characterized by high dimensionality, strong correlations, nonlinear interactions, and heterogeneous structure, which together give rise to rich phenomena such as multiscale organization, criticality, and emergent phases. Classical approaches to modeling and learning in these settings typically proceed by constructing reduced descriptions, approximate generative models, or sampling-based representations that aim to capture dominant correlations while discarding microscopic detail. In practice, these strategies rely on assumptions such as weak correlations, locality, separability, or efficient sampling from the underlying state or data distribution. While successful in many regimes, they can break down in strongly correlated, frustrated, or highly heterogeneous settings, where reduced descriptions become inaccurate, and sampling procedures suffer from slow mixing, large variance, or sensitivity to model assumptions. Quantum many-body systems represent an extreme instance of these challenges, as the exponential growth of Hilbert space with system size renders exact simulation infeasible and exposes fundamental limitations of classical approximation and Monte Carlo methods [1, 2]. Related difficulties also arise in data-driven domains such as healthcare and Earth system science, where data are noisy, heterogeneous, and often non-IID, complicating efforts to construct reliable models from limited observations [3–5].

Quantum machine learning (QML) has emerged as an interdisciplinary research area at the intersection of quantum information science and modern learning theory, motivated by the possibility that quantum resources can offer new approaches to representation, sampling, and inference in complex learning tasks [6–8]. Early visions of QML were closely tied to the prospect of quantum speedups for canonical machine learning primitives, including linear algebra routines, principal component analysis, clustering, and support vector machines, often relying on amplitude encoding and algorithmic subroutines such as phase estimation and Hamiltonian simulation. These works established important theoretical benchmarks, demonstrating that quantum computers could, under specific data access models, offer polynomial or exponential improvements in computational complexity. At the same time, they exposed practical challenges related to data loading, noise sensitivity, and error accumulation, which complicate the direct realization of such advantages on near-term quantum hardware [7–9]. As the field matured, the focus of QML shifted from purely algorithmic speedups toward the design of learning models and workflows compatible with noisy intermediate-scale quantum (NISQ) devices. This transition gave rise to hybrid quantum–classical approaches, in which parameterized quantum circuits or quantum subroutines are embedded within classical optimization loops and evaluated under realistic resource constraints [9, 10]. Within this paradigm, QML is no longer viewed as a single algorithmic framework but rather as a collection of complementary modeling strategies that use quantum resources for representation, optimization, or sampling in different ways.

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Beyond algorithmic design, the practical deployment of quantum machine learning algorithms also faces several hardware and systems-level challenges. Near-term quantum processors are affected by noise, decoherence, and limited circuit depth, which constrain the complexity of implementable learning models. Efficient state preparation, measurement overhead, and device connectivity further influence the scalability of hybrid quantum–classical workflows. Recent studies have therefore emphasized the importance of error mitigation, hardware-aware algorithm design, and resource-efficient circuit compilation for realizing QML on NISQ platforms. In addition, advances in quantum optimal control have provided powerful techniques for designing time-dependent control pulses that steer multi-qubit systems toward desired states or gate operations with high fidelity. Such control frameworks play an important supporting role in enabling the reliable implementation of quantum learning algorithms and quantum simulation workflows on emerging hardware.

Against this backdrop, QML has evolved into a diverse collection of paradigms, including variational quantum algorithms, quantum kernel methods, neural-network quantum states, and learning-theoretic analyses of trainability and generalization. These approaches have been explored across a growing range of scientific applications, including quantum many-body physics, materials science, drug discovery, biomedical data analysis, and climate modeling [11, 12]. This review surveys recent progress in applying quantum machine learning to complex systems and examines how different QML approaches address challenges related to representation, sampling, optimization, and deployment. We also highlight emerging theoretical tools for understanding the learning dynamics of quantum models and discuss application-driven frameworks in scientific and data-intensive domains.

This review surveys avenues in quantum machine learning for complex systems. We focus on a set of closely related approaches that address practical challenges in training, understanding, and deploying QML models. The remainder of the review is organized as follows. In Section 2, we survey the major paradigms in quantum machine learning, including variational quantum algorithms, quantum kernel methods, and neural-network quantum states, and discuss how these frameworks address representation and optimization challenges in complex quantum systems. Section 3 examines the learning dynamics and trainability of QML models, with particular emphasis on information propagation and scrambling-inspired diagnostics such as out-of-time-order correlators that provide insight into optimization landscapes. In Section 4, we review generative modeling approaches in QML, focusing on neural-network quantum states and hybrid quantum–classical frameworks such as quantum-enabled variational Monte Carlo for improving sampling efficiency in strongly correlated systems. Section 5 discusses distributed quantum learning paradigms, including federated quantum machine learning, which enables collaborative and privacy-preserving models training across decentralized datasets and heterogeneous quantum–classical infrastructures. Finally, Section 6 highlights emerging applications of QML to complex systems in domains such as drug discovery, cancer biology, and agro-climate modeling, where high-dimensional and structured data present significant challenges for classical learning methods.

2. Major paradigms in quantum machine learning

2.1. Variational circuit-based learning

The emergence of noisy intermediate-scale quantum (NISQ) devices catalyzed a transition from fully quantum learning algorithms toward hybrid quantum–classical frameworks. Within this paradigm, variational quantum algorithms (VQAs) formulate learning, simulation, and optimization tasks as minimization problems over parameterized quantum circuits (PQCs), with classical optimizers updating circuit parameters based on measurement outcomes [10]. This approach enabled some of the earliest practical demonstrations of quantum-assisted learning and simulation, most notably through the variational quantum eigensolver (VQE), which approximates ground-state properties of quantum systems using shallow circuits and iterative optimization [13].

Building on this framework, quantum neural networks (QNNs) were introduced as circuit-based analogues of classical neural networks, with tunable gates acting as trainable parameters. QNNs have been explored for supervised learning, generative modeling, and quantum data processing, and have been implemented on both simulators and available quantum hardware. However, extensive theoretical and numerical studies have revealed that many variational circuit architectures suffer from barren plateaus, where gradients vanish exponentially with system size or circuit depth, rendering training ineffective [14]. Subsequent work has investigated how circuit expressivity, entanglement structure, and connectivity influence gradient statistics, highlighting fundamental trade-offs between representational power and trainability [15]. These insights have motivated the development of structured ansätze, locality-preserving circuits, problem-informed initialization strategies, and alternative optimization techniques, although scalability remains an open challenge in highly correlated or rugged learning landscapes.

2.2. Kernel- and embedding-based quantum learning

Quantum kernel methods provide an alternative approach to QML that avoids explicit variational training of quantum parameters. In this paradigm, quantum circuits define implicit feature maps that embed data into high-dimensional Hilbert spaces, while learning is performed through kernel evaluation and classical optimization [16, 17]. Quantum kernels can induce complex decision boundaries using relatively shallow circuits, making them attractive for learning tasks in low-data or structured regimes.

At the same time, the practical performance of quantum kernels depends critically on the efficiency and accuracy of kernel estimation, which can be affected by noise, shot statistics, and circuit depth. Recent studies have investigated the expressivity, generalization properties, and classical simulability of quantum kernels, clarifying the regimes in which they may offer advantages over classical counterparts. This body of work suggests that quantum kernel methods are most effective when the structure of the feature map aligns closely with the underlying data distribution, rather than as universal replacements for classical kernel techniques.

2.3. Neural-network quantum states

Neural-network quantum states (NQS) constitute a representational paradigm within QML in which quantum many-body wavefunctions are parameterized directly by classical machine learning models. The seminal demonstration that restricted Boltzmann machines (RBMs) can approximate ground states of interacting spin systems initiated a broad research program on neural-network-based wavefunction representations [18]. Subsequent theoretical and numerical studies extended these ideas to frustrated magnets, fermionic systems, topological phases, and models with long-range interactions, establishing NQs as flexible variational ansätze for strongly correlated quantum matter [19–21]. NQs combine ideas from variational Monte Carlo, statistical mechanics, and representation learning, providing a compact way to encode high-order correlations and entanglement. A wide range of architectures, including deep neural networks, convolutional models, and autoregressive constructions, have been proposed to enhance expressivity and sampling efficiency. Despite their flexibility, NQS-based methods typically rely on Monte Carlo estimation of observables and gradients, making training performance sensitive to autocorrelation times, mixing efficiency, and estimator variance, particularly in systems with rugged or multi-modal configuration spaces.

2.4. Distributed and hybrid quantum–classical learning frameworks

Many quantum machine learning algorithms are implemented within hybrid quantum–classical architectures, where quantum circuits are embedded inside classical optimization loops. In these frameworks, quantum hardware is typically used to prepare quantum states or evaluate expectation values, while classical processors perform parameter updates, gradient estimation, and data preprocessing. Such hybrid workflows form the basis of many near-term QML algorithms, including variational quantum circuits and quantum kernel methods, and are designed to operate within the constraints of noisy intermediate-scale quantum (NISQ) devices [10, 22].

More recently, distributed quantum learning paradigms have been proposed in which training tasks are shared across multiple quantum processors or computational nodes. These approaches aim to overcome hardware limitations and enable collaborative training across distributed datasets, for example through cloud-based hybrid workflows or federated quantum learning systems where local models are trained on decentralized data and only limited updates are exchanged. Such frameworks are particularly relevant for applications involving large datasets or privacy-sensitive environments, and are discussed in more detail in Section 5.1.

Table 1 summarizes the major paradigms in quantum machine learning (QML).

3. Learning dynamics and trainability in quantum models

The trainability of quantum machine learning (QML) models is largely governed by the structure of their optimization landscapes. Most QML algorithms are formulated as variational optimization problems in which a parameterized quantum state $|\psi(\theta)\rangle$ or circuit $U(\theta)$ is trained by minimizing a cost function defined through expectation values of quantum observables. As a result, understanding the geometry of the resulting loss landscapes has become a central theme in recent QML research.

Several studies have analyzed the statistical properties of these landscapes and their implications for optimization. Early work on variational quantum circuits showed that highly expressive parameterizations can lead to landscapes that resemble random functions over Hilbert space, making gradient-based optimization challenging [14]. Subsequent investigations explored how circuit structure, initialization strategies, and cost-function locality influence the landscape geometry and the efficiency of gradient estimation [10, 22, 23]. Related work has examined the relationship between ansatz expressibility, entanglement generation, and optimization difficulty [24, 25].

Table 1 • Comparison of major paradigms in quantum machine learning.

Paradigm	Data type	Core primitive	Bottlenecks	Application regimes	Hardware requirements
VQA/QNN	Quantum states or encoded classical data	Parameterized quantum circuits with hybrid optimization	Barren plateaus, shot noise, and circuit depth limitations	Variational quantum simulation and supervised learning	NISQ devices with moderate qubit counts and shallow circuits
Quantum kernel methods	Classical data in quantum feature space	Quantum feature maps and kernel evaluation	Noise in kernel estimation, sampling cost, and limited scalability	Low-data and structured classification problems	Shallow NISQ circuits for state overlap estimation
Neural-network quantum states (NQS)	Quantum many-body wavefunctions	Neural-network ansatz with variational Monte Carlo	Sampling inefficiency and gradient variance	Strongly correlated many-body systems, ground states, and dynamics	Classical compute with optional quantum sampling
Tensor-network methods	Structured quantum many-body states	Low-rank tensor decompositions: MPS, PEPS, and MERA	Tensor contraction cost in higher dimensions	1D and area-law-entangled systems	Classical high-performance computing
Federated QML	Distributed classical or quantum datasets	Privacy-preserving distributed training	Communication overhead and data heterogeneity	Multi-institution and privacy-sensitive learning settings	Networked hybrid quantum–classical infrastructure

VQA: Variational Quantum Algorithms; QNN: Quantum Neural Networks; NQS: Neural Network Quantum States; QML: Quantum Machine Learning; MPS: Matrix Product States; PEPS: Projected Entangled Pair States; MERA: Multiscale Entanglement Renormalization Ansatz; NISQ: Noisy Intermediate-Scale Quantum.

In parallel, optimization landscapes have also been studied in the context of neural-network quantum states (NQS), where the variational ansatz is represented by classical machine learning architectures such as restricted Boltzmann machines, convolutional networks, and autoregressive models [18, 26, 27]. In these systems, the geometry of the optimization landscape is influenced not only by the expressive capacity of the model but also by the sampling dynamics of variational Monte Carlo algorithms. Consequently, insights from statistical physics, such as energy landscape theory and correlation structure in many-body systems, have increasingly been applied to analyze learning behavior in quantum models [28].

3.1. Barren plateaus and gradient concentration

A major obstacle in the training of quantum machine learning models is the emergence of barren plateaus, regions of the optimization landscape where gradients become exponentially small as the system size increases. In such regimes, estimating gradients with sufficient precision requires an exponentially large number of measurements, rendering gradient-based optimization impractical for large-scale systems. This phenomenon was first systematically analyzed in the context of variational quantum circuits by McClean et al. [14], who showed that for sufficiently expressive parameterized circuits, the variance of the gradient of the cost function decays exponentially with the number of qubits.

More precisely, consider a parameterized quantum circuit $U(\theta)$ with parameters θ , and a cost function defined as the expectation value of an observable (H),

$$C(\theta) = \langle 0 | U^\dagger(\theta) H U(\theta) | 0 \rangle. \quad (1)$$

McClean et al. [14] showed that for random parameterized circuits approaching approximate unitary 2-designs, the variance of the

gradient of the cost function obeys

$$\text{Var}[\partial_{\theta_i} C(\theta)] \sim \mathcal{O}(b^{-n}), \quad (2)$$

where n is the number of qubits and $b > 1$ is a constant depending on the circuit architecture. Consequently, the typical magnitude of the gradient decreases exponentially with system size. This behavior is illustrated in **Figure 1**, which shows the exponential decay of gradient variance as the number of qubits increases. The result can be understood as a manifestation of concentration of measure in high-dimensional Hilbert spaces: when circuits become sufficiently random, expectation values concentrate sharply around their mean values, leaving little information in the gradient signal to guide optimization.

The underlying geometric intuition for barren plateaus is illustrated in **Figure 1**. As the dimension of Hilbert space grows exponentially with system size, randomly parameterized quantum states become almost uniformly distributed over the state manifold. In this regime, the cost function becomes nearly constant over large regions of parameter space, resulting in extremely flat optimization landscapes.

Subsequent work has refined this picture by showing that the severity of barren plateaus depends strongly on the structure of the cost function and the depth of the circuit. In particular, Cerezo et al. demonstrated that global cost functions, which depend on observables acting on the entire system, exhibit exponentially vanishing gradients even for relatively shallow circuits. In contrast, local cost functions involving observables acting on only a few qubits may retain polynomially scaling gradient variance in certain regimes [23]. This distinction is illustrated in **Figure 2**, where the onset of barren plateaus occurs much earlier for global objectives than for local ones.

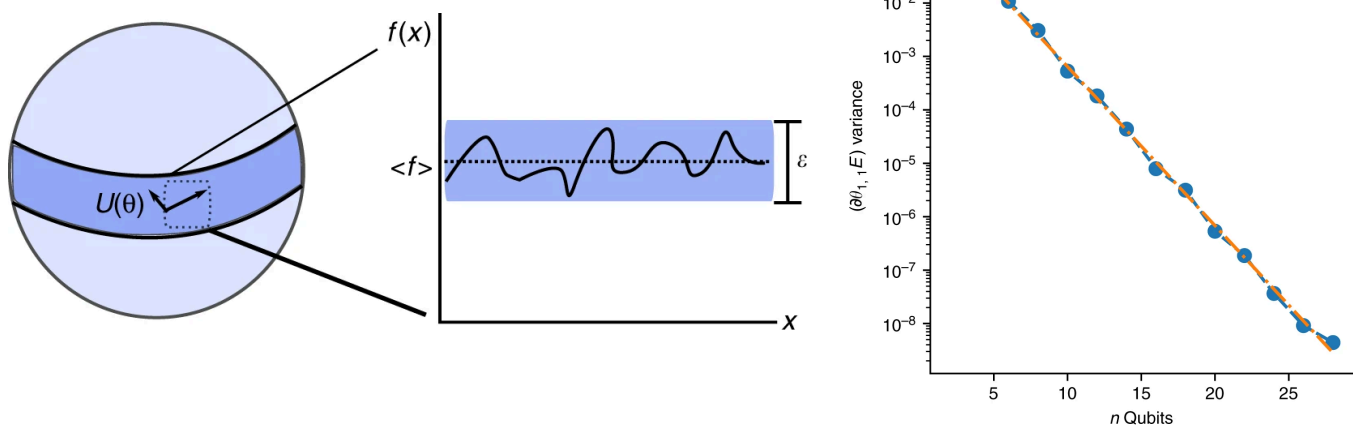


Figure 1 • (Left) Illustration of concentration of measure in high-dimensional Hilbert space. As circuit parameters explore the state manifold, expectation values of observables concentrate around their mean values, leading to flat optimization landscapes. **(Right)** Exponential decay of the variance of the gradient with the number of qubits, demonstrating the barren plateau phenomenon in random variational circuits. Reproduced from [14].

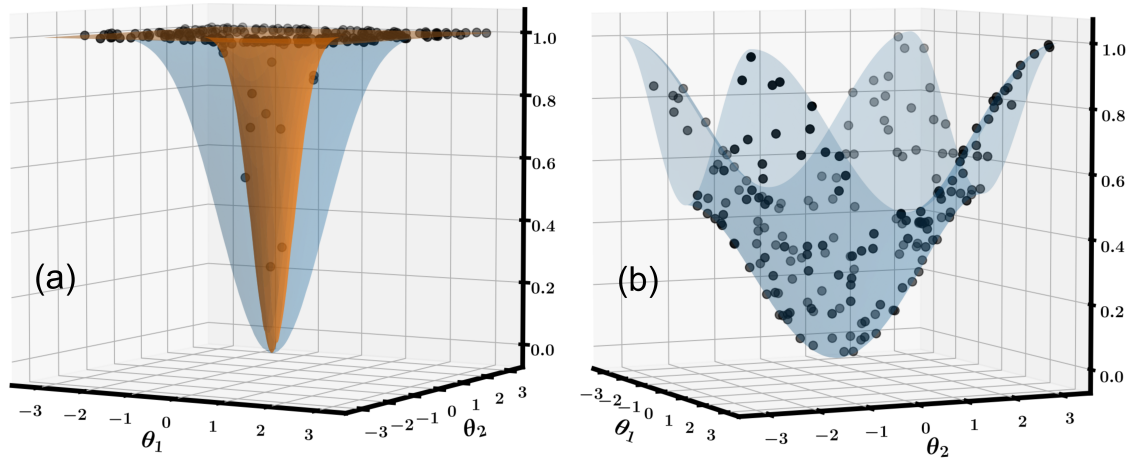


Figure 2 • Dependence of barren plateau behavior on cost-function locality. (a) Global cost functions lead to exponentially suppressed gradients even in shallow circuits, while (b) local cost functions can remain trainable over a larger range of circuit depths. Reproduced from [23].

The depth of the quantum circuit also plays a crucial role in determining trainability. As shown in **Figure 3**, increasing circuit depth tends to increase the expressibility of the ansatz, but also accelerates the onset of barren plateau behavior. When circuits become sufficiently deep to approximate unitary 2-designs, gradient concentration emerges generically. Consequently, many modern QML architectures emphasize structured or problem-inspired ansätze that limit circuit expressibility in order to maintain meaningful gradient signals.

While barren plateaus represent a fundamental limitation of highly expressive quantum circuits, subsequent work has shown that they do not exhaust the trainability challenges faced by QML models. Even in regimes where gradients remain finite, optimization landscapes may still contain numerous traps and poor local minima that hinder efficient training [29]. Conversely, certain structured architectures, such as quantum convolutional neural networks, have been shown to avoid barren plateaus entirely due to their hierarchical and locality-preserving design [30].

These results highlight that trainability in QML depends sensitively on the interplay between circuit architecture, cost-function

design, and system size. As a result, a growing body of research has focused on identifying ansätze and optimization strategies that preserve gradient signal while maintaining sufficient expressive power. Understanding this balance remains a central challenge in the development of scalable quantum machine learning algorithms.

3.2. Information propagation and dynamical diagnostics

A central challenge in quantum machine learning (QML) is understanding the structure and navigability of the underlying loss landscape. Trainability issues, such as barren plateaus, highly non-uniform gradients, and sensitivity to initialization, are often closely tied not only to the magnitude of gradients but also to how information propagates through the model during optimization. In classical deep learning, the propagation of signals through network layers strongly influences trainability and generalization. Analogous questions arise in QML, where correlations and entanglement determine how local perturbations influence the global quantum state.

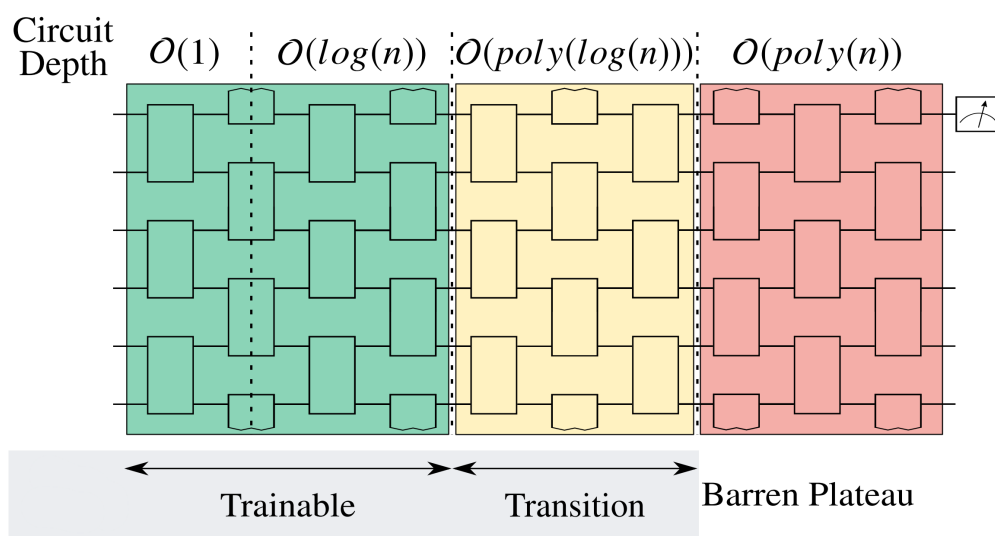


Figure 3 • Influence of circuit depth on the emergence of barren plateaus. Increasing circuit depth increases expressibility but can also drive the circuit toward regimes where gradients vanish exponentially. Reproduced from [23].

Recent work has therefore emphasized the relationship between trainability and information spreading in quantum systems. In the context of variational circuits, analyses of entanglement growth and operator spreading have been used to characterize how rapidly quantum circuits scramble information [31]. These studies suggest that excessive information scrambling can suppress useful optimization signals and contribute to barren plateau behavior, whereas structured ansätze that preserve locality or controlled entanglement growth may retain more trainable landscapes. More broadly, tools from quantum many-body physics, such as entanglement entropy, correlation functions, and dynamical diagnostics of operator growth, have begun to play an increasingly important role in analyzing learning dynamics in quantum models [32, 33].

Among these probes, out-of-time-order correlators (OTOCs) have emerged as a particularly useful diagnostic of information propagation in quantum systems. Originally introduced to characterize operator spreading and scrambling, OTOCs quantify how strongly a local perturbation at one part of a system influences distant degrees of freedom as the system evolves. In a recent work [34], it is shown that the imaginary component of the OTOC, a quantity historically overlooked in both physics and QML contexts, encodes detailed geometric information about the learning dynamics of a graph-based neural-network quantum state.

A restricted Boltzmann machine (RBM) is chosen as a learner, which can be viewed as a bipartite network of visible spins $\{v_i\}$ and hidden spins $\{h_j\}$. The trainable parameters $X = (a, b, W)$ consist of visible biases a_i , hidden biases b_j , and weights W_{ij} connecting the two layers. These parameters define an effective Ising-type Hamiltonian

$$H(X) = \sum_i a_i \sigma_z(v_i) + \sum_j b_j \sigma_z(h_j) + \sum_{i,j} W_{ij} \sigma_z(v_i) \sigma_z(h_j), \quad (3)$$

whose associated thermal state

$$\rho_{\text{th}}(X) = \frac{e^{-H(X)}}{\text{Tr}[e^{-H(X)}]} \quad (4)$$

generates the probability distribution underlying the RBM ansatz.

To make this idea concrete, the response of two local operators associated with visible and hidden neurons is considered, and how their non-commutativity evolves under the dynamics generated by the learner Hamiltonian is tracked.

In this setting, U_1 and U_2 are chosen to act on a visible neuron v_k and a hidden neuron h_m . The OTOC then takes the form

$$C_{\alpha,\beta}(X, t) = \langle \tilde{\sigma}^\alpha(v_k) \tilde{\sigma}^\beta(h_m, t) \tilde{\sigma}^\alpha(v_k) \tilde{\sigma}^\beta(h_m, t) \rangle_{\rho_{\text{th}}(X)}, \quad (5)$$

where $\alpha, \beta \in \{x, y\}$ and $\tilde{\sigma}$ denotes a shifted Pauli operator, and the time evolution of operators is generated by $H(X)$.

Theorem II.1 of Ref. [34] shows that, for $\kappa_1 = \kappa_2 = 0$ and $H_{\text{OTOC}} = H(X)$, the OTOC admits a closed analytic form

$$C_{\alpha,\beta}(X, \tau) = \cos(\tau) + i \langle \sigma_z(v_k) \sigma_z(h_m) \rangle_{\rho_{\text{th}}(X)} \sin(\tau), \quad \tau = 4W_{km}t, \quad (6)$$

and obeys a family of invariants of motion constructed from its real and imaginary parts.

Equation (6) shows that the imaginary part of the OTOC is directly proportional to the visible–hidden correlation

$\langle \sigma_z(v_k) \sigma_z(h_m) \rangle$. This observation motivates defining a covariance-like quantity $\eta(X) = \text{Cov}(\sigma_z(v_k), \sigma_z(h_m))$, which captures the strength of correlations between visible and hidden neurons. In Ref. [34], it was shown that this quantity can be reconstructed directly from appropriately shifted OTOCs evaluated at a specific time, thereby establishing a direct link between dynamical OTOC measurements and static correlation properties of the learner.

Having related the imaginary part of the OTOC to pairwise correlations, next this quantity is connected to standard information-theoretic measures, in particular the quantum mutual information $I(v_k, h_m)$ between the visible neuron v_k and the hidden neuron h_m . From the thermal state (**Equation (4)**) one can construct the one- and two-spin reduced density matrices $\rho^{(1)}(v_k)$, $\rho^{(1)}(h_m)$, and $\rho^{(2)}(v_k, h_m)$, and hence the mutual information

$$I(v_k, h_m) = S(\rho^{(1)}(v_k)) + S(\rho^{(1)}(h_m)) - S(\rho^{(2)}(v_k, h_m)), \quad (7)$$

where $S(\rho) = -\text{Tr}(\rho \ln \rho)$ is the von Neumann entropy. Theorem II.2 of Ref. [34] establishes that, for the RBM learner, $I(v_k, h_m)$ and $\eta(X)$ are constrained to lie inside a convex region of the (I, η) -plane bounded by upper and lower bounds as depicted in **Figure 4**.

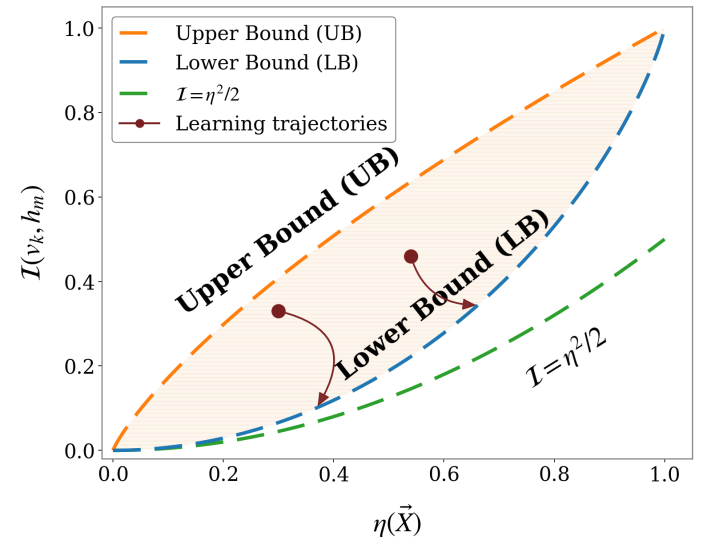


Figure 4 • The figure shows the analytic bounds in the (I, η) space. The lower bound (LB) is $\text{LB}(\eta) = 2 - 2\ell\left(\frac{1+\eta}{4}\right) - 2\ell\left(\frac{1-\eta}{4}\right)$ and upper bound (UB) is $\text{UB}(\eta) = \ell\left(\frac{1}{2} + \frac{1}{2}\sqrt{1-|\eta|}\right) + \ell\left(\frac{1}{2} - \frac{1}{2}\sqrt{1-|\eta|}\right)$ for the RBM learner, where $\ell(x) = -x \ln x$. Provided alongside in green is the conventionally known lower bound for arbitrary bipartitions [35]. Representative learning trajectories of the network G during training are shown within the allowed region. Adapted from [34]. RBM: Restricted Boltzmann Machine.

To build intuition for these bounds, it is helpful to interpret the (I, η) plane as a geometric constraint on the correlations that can develop between visible and hidden units during training. The quantity η measures pairwise correlations extracted from the imaginary component of the OTOC, while $I(v_k, h_m)$ quantifies the total information shared between the two units. The analytic bounds, therefore, restrict how rapidly information can accumulate as correlations grow. During optimization, parameter updates move the learner through this region, producing trajectories that reflect how correlations are redistributed across the network. Due

to the existence of the bounds, the pair $(I(v_k, h_m), \eta(X))$ for every visible–hidden pair thus lives in an emergent I – η space, and the training dynamics of the learner can be visualized as trajectories within this convex region. **Figure 4** illustrates typical learning trajectories in this space during training of the network, together with the analytic bounds. It must be noted that the lower bound is strictly tighter than previously known generic bounds of the form $I \geq \eta^2/2$ for arbitrary bipartite systems.

To explore the structure of this emergent I – η space, the neural-network quantum state is trained to approximate the ground state of a driver Hamiltonian H_{drv} using stochastic re-configuration (natural-gradient) optimization of the network parameters X [36]. In this framework, the RBM wave function $\psi(v, X)$ defines a probability distribution $|\psi(v, X)|^2$ over spin configurations, from which observables and correlation measures can be estimated through Monte Carlo sampling.

The framework is benchmarked on two families of driver Hamiltonians: the transverse-field Ising model (TFIM) with nearest-neighbor couplings, and a concentric TFIM (c-TFIM) with long-range “concentric” couplings that generate volume-law entangled ground states. Both share the generic form

$$H_{\text{drv}} = -B \sum_i \sigma_x(i) - \sum_{ij} J_{ij} \sigma_z(i) \sigma_z(j), \quad (8)$$

where J_{ij} encodes either nearest-neighbor interactions (TFIM) or long-range concentric couplings (c-TFIM), and the ratio $g = B/J_0$ controls the phase of the system.

After training, the correlation structure between visible and hidden neurons can be analyzed by computing the corresponding points in the I – η plane for all visible–hidden pairs. This representation allows the learning dynamics of the RBM to be interpreted in terms of the information-theoretic bounds discussed above.

Several notable features emerge from this analysis. First, for all system sizes N and values of the control parameter g , the trained RBM representations lie on the analytic lower bound $\text{LB}(\eta)$ in the I – η plane. This indicates that, for a given covariance $\eta(X)$ between visible and hidden spins, the RBM organizes correlations so as to minimize the mutual information $I(v_k, h_m)$ within the allowed convex region. This lower-bound saturation appears robust across different driver models, suggesting a general learning principle: the network represents the target quantum state using the minimal mutual information compatible with the required covariance structure.

Second, the distribution of points in the I – η space reflects the correlation structure of the underlying physical system. For strongly correlated regimes ($g \rightarrow 0$), the trained networks exhibit large values of both $\eta(X)$ and $I(v_k, h_m)$, while in the product-state limit ($g \rightarrow \infty$) the distribution collapses toward the origin $(I, \eta) = (0, 0)$. This behavior indicates that the RBM implicitly learns the correlation structure of the driver Hamiltonian, even though no prior information about the physical system is provided during training.

Finally, the I – η representation distinguishes different entanglement structures in the driver models. For the nearest-neighbor TFIM, the distribution of points remains relatively narrow for fixed g , reflecting a more uniform allocation of correlations between visible and hidden units. In contrast, for the fully connected TFIM with volume-law entanglement, the networks display a

broader spread of (I, η) values, particularly near $g \rightarrow 0^+$, indicating multiple possible assignments of correlations compatible with the same learned state.

Taken together, these results demonstrate that the imaginary component of the OTOC, and the associated I – η representation, provides a dynamical probe of the learning landscape of quantum machine learning models. This framework links the scrambling of information within the learner to physical properties of the target system, such as phase transitions, correlation ranges, and entanglement structure, suggesting that tools originally developed to study quantum chaos may also provide insight into the trainability of QML architectures for complex quantum systems.

RBMs as an analytical testbed for QML theory: The analytical results presented above rely on the specific structure of restricted Boltzmann machines (RBMs), which provide a particularly convenient theoretical setting for studying neural-network quantum states. Although RBMs are no longer the dominant architecture in modern classical machine learning, where deeper networks and attention-based models are more common, their bipartite structure admits compact analytical representations of the wave function. After marginalizing over hidden variables, correlations and probability amplitudes can often be written in closed form, making RBMs well suited for investigating fundamental questions about expressibility, optimization dynamics, and information flow in quantum variational states.

In the present analysis, the RBM structure allows the learner Hamiltonian to be expressed in a simple Ising form, enabling closed expressions for the out-of-time-order correlator (OTOC) and the associated covariance and mutual-information bounds. Such analytic access is typically unavailable for deeper neural-network ansätze, where the effective probability distributions and Hamiltonians become considerably more complex. In this sense, RBMs provide a mathematically transparent testbed in which information-theoretic diagnostics of learning dynamics can be derived rigorously.

While the exact relations derived here rely on the RBM architecture, the underlying conceptual framework is more general. Many modern neural-network quantum states, including convolutional, autoregressive, and transformer-based architectures, also represent correlations through parameterized interactions that can be interpreted as effective many-body couplings. For these more expressive models, quantities such as OTOCs, covariance measures, or mutual-information bounds would typically need to be estimated numerically rather than obtained in closed form, but they may still provide valuable diagnostics of correlation structure, information flow, and trainability. From this perspective, RBMs should not be viewed as a final architecture for large-scale machine learning, but rather as a controlled theoretical platform for developing tools that may later be applied to more expressive neural-network quantum states.

4. Generative models in quantum machine learning

Generative modeling occupies a central place in quantum machine learning because many learning tasks in quantum systems ultimately reduce to representing, sampling, or manipulating high-dimensional probability distributions and wave functions. In the context of many-body physics, one often seeks compact representations of quantum

states whose amplitudes, phases, or reduced measurement statistics can be learned from data or optimized variationally. This perspective has made generative models a natural bridge between modern machine learning and quantum state representation, with neural-network quantum states (NQS) emerging as one of the most successful realizations of this idea [18, 28, 37].

A useful viewpoint in quantum machine learning is to regard a quantum state as defining a structured probability model over measurement outcomes. In variational approaches, one introduces a parameterized state $\psi_\theta(x)$, where x denotes a basis configuration, and trains the parameters θ so that the resulting distribution $|\psi_\theta(x)|^2$ or related observable statistics approximate a target quantum state or minimize a physical cost function [18, 38, 39]. From this perspective, variational wave-function learning shares many conceptual features with classical generative modeling, including representation learning, likelihood-free inference, and sampling from high-dimensional structured distributions.

This perspective has proven useful across a range of settings. In variational Monte Carlo, the generative model is trained indirectly through estimates of local energies and gradients [18, 40]. In quantum state tomography, neural generative models can be trained directly from measurement data to reconstruct the unknown state [38, 41]. Related ideas have also been extended to mixed states and open quantum systems through neural density operators and purification-based ansätze [38, 39, 42]. These developments position generative modeling not merely as a convenient analogy, but as a unifying language for representing and learning quantum states across equilibrium, dynamical, and data-driven tasks.

4.1. Neural-network quantum state architectures

The earliest and most influential example of a neural-network quantum state was the restricted Boltzmann machine (RBM), introduced as a variational ansatz for many-body wave functions by Carleo and Troyer [18]. Since then, the architectural landscape of NQSs has expanded substantially. Deep Boltzmann machines, feedforward neural networks, convolutional networks, recurrent architectures, and autoregressive models have all been explored as wave-function representations [26, 27, 43–46]. A schematic

diagram of a Restricted Boltzmann machine has been shown in **Figure 5**.

Different architectures provide different inductive biases. Convolutional NQSs are naturally suited to lattice systems with translation symmetry and locality [26, 47]. Recurrent and autoregressive ansätze support direct sampling and can avoid some of the mixing bottlenecks associated with Markov-chain methods [27, 44]. **Figure 6** shows two popular Autoregressive NQS models, namely, Recurrent NQS and Convolutional NQS. More recently, transformer-inspired and attention-based constructions have begun to appear in the NQS literature, motivated by their ability to capture long-range dependencies and flexible correlation patterns [47–49]. **Figure 7** shows the architecture of a Transformer neural quantum state, and **Figure 8** shows the performance of Transformer NQS on the Transverse Field Ising Model. Together, these developments suggest that NQSs should be viewed not as a single model family, but as a broad architectural class in which different neural inductive biases can be adapted to different quantum learning tasks.

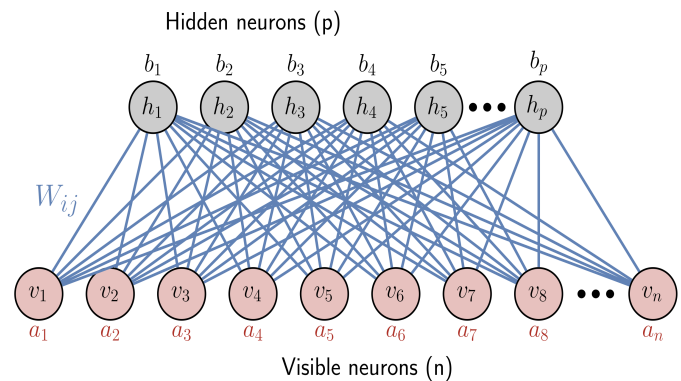


Figure 5 • Restricted Boltzmann machine neural quantum state. The wave function is represented using a bipartite network consisting of visible units corresponding to physical degrees of freedom and hidden units that capture correlations between them. The parameters of the model include visible and hidden biases and the weights connecting the two layers. Reproduced from [50].

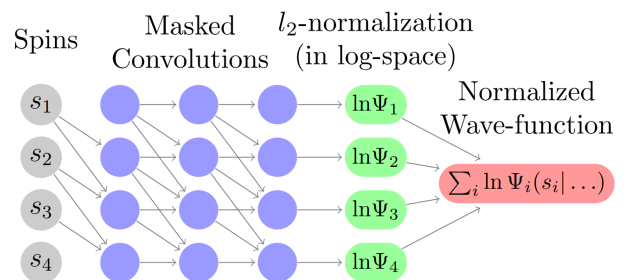
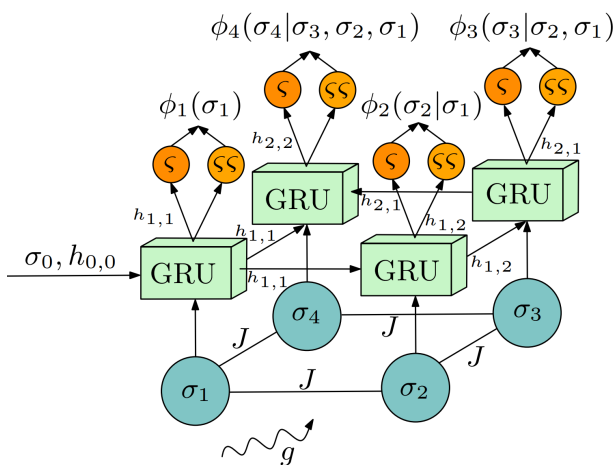


Figure 6 • Autoregressive neural-network quantum state (NQS) architectures. **(Left)** Recurrent NQS, where spin configurations are processed sequentially using recurrent units, generating conditional amplitudes for each spin based on previously generated configurations. Reproduced with permission from [51]. **(Right)** Convolutional autoregressive NQS using masked convolutions, which enforce the autoregressive factorization and enable efficient direct sampling of many-body configurations. Reproduced with permission from [27]. GRU: Gated Recurrent Unit.

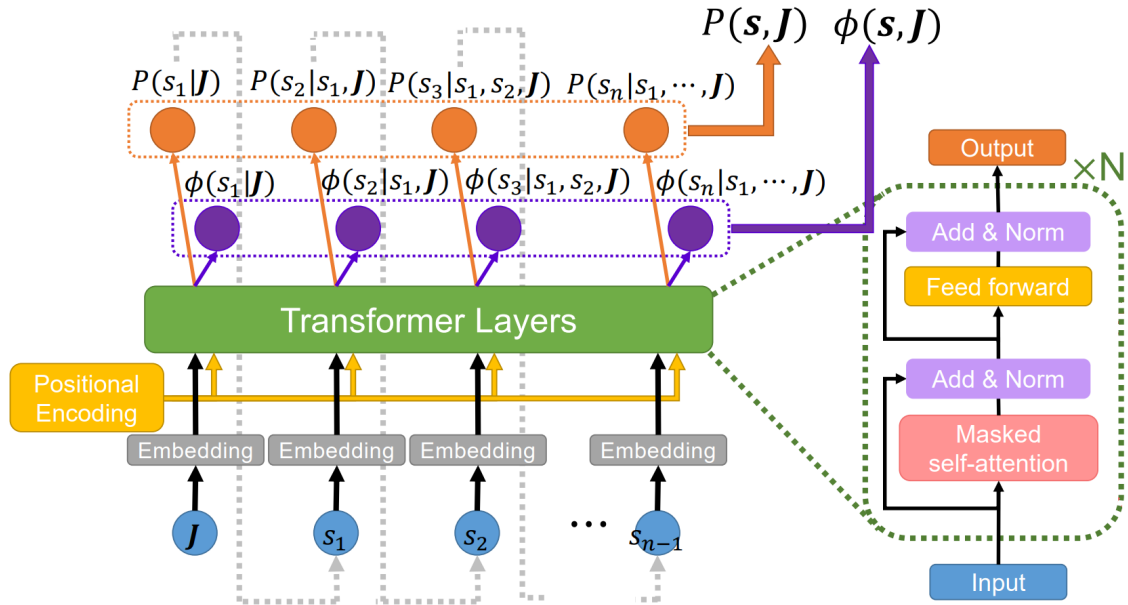


Figure 7 • Transformer neural quantum state architecture. Spin configurations are embedded and processed through a stack of transformer layers with masked self-attention. The autoregressive structure ensures that the conditional amplitudes are generated sequentially, allowing the wave function to be expressed as a product of conditional probabilities. Reproduced with permission from [52].

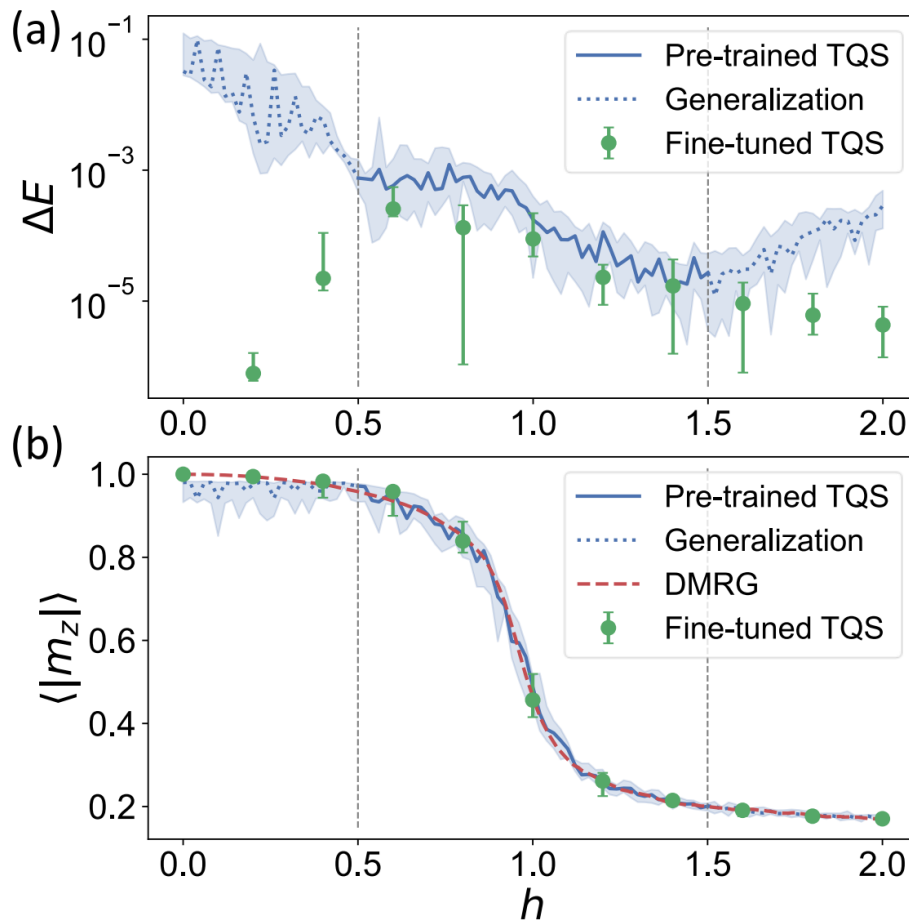


Figure 8 • Performance of transformer neural quantum states for the transverse-field Ising model. (a) Ground-state energy error (ΔE) as a function of transverse field (h). The pre-trained transformer quantum state (TQS) shows strong performance within the training regime, but generalizes poorly outside it. Fine-tuning significantly improves accuracy across the full range of (h). (b) Magnetization ($\langle |m_z| \rangle$) as a function of (h), showing excellent agreement between TQS predictions and DMRG results across the phase transition. Fine-tuned TQS closely matches the reference curve, capturing the critical behavior. Reproduced with permission from [52]. DMRG: Density Matrix Renormalization Group.

Expressibility and representation power of NQs: A central theoretical question is how expressive neural-network quantum states are as representations of many-body wave functions. Several works have established that RBMs and related architectures are universal approximators for broad classes of probability distributions and can efficiently represent certain highly entangled states that are difficult to capture with more traditional ansätze [43, 53, 54]. Other analyses have investigated the relationship between NQs and tensor-network states, showing that while some tensor-network families can be embedded efficiently within neural-network representations, the converse need not hold in a straightforward way [27, 43, 55].

Expressibility, however, is not only a question of existence but also of practical learnability. A model may be expressive in principle while remaining difficult to train because of poor optimization geometry, unfavorable sampling complexity, or inefficient parameter scaling. Recent work has therefore examined expressibility together with trainability, entanglement structure, and the scaling of representation error [19, 28, 56]. These studies suggest that the usefulness of an NQS architecture depends on a balance between representational flexibility and algorithmic accessibility, rather than on expressibility alone.

Applications of neural-network quantum states: Neural-network quantum states have been applied across a wide range of problems in quantum physics and chemistry. In equilibrium settings, they have been used extensively for ground-state searches in spin models, frustrated magnets, fermionic systems, and quantum chemistry Hamiltonians [18, 26, 45, 57]. In these applications, NQs provide compact variational ansätze that can encode complex correlation structures beyond those captured by simple mean-field or low-entanglement approximations.

Beyond equilibrium physics, NQs have also been extended to quantum dynamics, open-system evolution, and state reconstruction from measurement data. Time-dependent variational principles and autoregressive neural states have enabled the simulation of real-time and dissipative dynamics in interacting many-body systems [44, 51, 58, 59]. Neural quantum state tomography and neural density operator approaches have likewise broadened the scope of these models to mixed states and experimentally relevant learning tasks [38, 39]. Collectively, these applications illustrate that NQs serve not only as compact representations of wave functions, but also as flexible computational tools for learning and simulating complex quantum systems.

4.2. Challenges in variational Monte Carlo

Despite their expressive power, many NQS approaches remain limited in practice by the difficulty of sampling from the distributions they define. In variational Monte Carlo, expectation values and gradients are typically estimated from configurations sampled according to $|\psi_\theta(x)|^2$, or an associated importance distribution [18, 40]. For generic NQs, direct sampling is often unavailable, and one instead relies on Markov chain Monte Carlo (MCMC) schemes such as Metropolis–Hastings updates.

This introduces a substantial algorithmic bottleneck. In rugged, frustrated, or multimodal landscapes, classical Markov chains may suffer from long autocorrelation times and slow mixing, degrading the statistical quality of gradient and energy estimates [37, 40]. As a result, the practical performance of an

NQS ansatz is shaped not only by its expressibility, but also by the quality of the sampling procedure used to explore its configuration space. This observation has motivated extensive research into alternative samplers, autoregressive constructions enabling direct generation of samples, and hybrid classical–quantum strategies for accelerating variational Monte Carlo workflows [26, 27, 44].

Quantum-assisted sampling strategies: The use of quantum resources to accelerate sampling has long been considered a promising direction for overcoming classical Monte Carlo bottlenecks. Quantum walks, quantum Gibbs-sampling protocols, and Hamiltonian-simulation-based methods have all been explored as possible routes toward improved sampling complexity in structured probability distributions [60–62]. In the context of variational algorithms, these ideas suggest that quantum devices may be particularly useful not only as variational ansätze, but also as samplers embedded within broader classical optimization loops.

More recent work has investigated hybrid strategies in which quantum circuits are used to generate proposal distributions or sampling subroutines that complement classical variational optimization [12, 50, 63]. This perspective is especially natural for NQS-based methods, where the primary computational bottleneck often lies in Monte Carlo estimation rather than in the representational form of the ansatz itself. The resulting class of methods sits at the interface of generative modeling, quantum simulation, and variational inference.

4.3. Quantum-enabled VMC algorithm for training NQs

Although NQs have proven remarkably expressive across a wide range of correlated quantum systems, their practical performance in large-scale simulations is often limited by the cost of classical sampling. Conventional variational Monte Carlo (VMC) optimization relies on classical Markov chain Monte Carlo (MCMC) sampling, whose mixing times and autocorrelation lengths can grow exponentially with system size. This challenge becomes especially severe in frustrated or multireference regimes, where the probability distribution encoded by the NQS wave function develops multiple deep local minima separated by high-energy barriers. As a result, classical samplers struggle to explore configuration space efficiently, making the optimization landscape rugged and slowing convergence. This sampling bottleneck represents one of the central obstacles to scaling NQS-based electronic structure and condensed matter simulations beyond modest system sizes.

In a recent work [50], a quantum-enabled variational Monte Carlo (Q-VMC) framework is introduced that replaces the classical sampling step with a shallow quantum circuit implementing the dynamics of an auxiliary Ising-like Hamiltonian designed to mimic the structure of the desired probability distribution. The key conceptual step is the construction of a surrogate Hamiltonian h_1 whose Gibbs distribution approximates the probability distribution of the NQS $|\psi_\theta(x)|^2$.

The surrogate Hamiltonian is written as

$$h_1 = \sum_i a_i(\vec{X})\sigma_z^{(i)} + \sum_{i<j} J_{ij}(\vec{X})\sigma_z^{(i)}\sigma_z^{(j)} \quad (9)$$

and $h_2 = \sum_i \sigma_x^{(i)}$ acts as a transverse-field mixer. The corresponding unitary evolution operator is

$$U(\tau, \gamma) = \exp[-i\tau(\gamma h_1 + (1 - \gamma)h_2)] \quad (10)$$

with $(\gamma, \tau) \in \mathbb{R}$. The parameters $a_i(\vec{X})$ and $J_{ij}(\vec{X})$ are learned variational parameters that determine the rotation angles of the single-qubit and two-qubit gates appearing in the quantum circuit implementation of $U(\tau, \gamma)$. The unitary dynamics is implemented through a first-order Trotter decomposition of the time propagator. The accuracy of the implemented proposal time propagator depends on the number of Trotter steps used to approximate the unitary evolution. Increasing the number of Trotter layers systematically improves the fidelity of the simulated dynamics and therefore the quality of the induced proposal distribution. However, deeper circuits also accumulate hardware noise, gate errors, and decoherence, which can degrade performance on near-term devices. In practice, one therefore chooses a modest number of Trotter steps that balances approximation accuracy with the noise limitations of available quantum hardware. A detailed analysis of the Trotterization error and circuit resource requirements is provided in Ref. [50]. Importantly, the correctness of the sampling procedure does not require an exact implementation of the proposal unitary. The quantum circuit only generates candidate configurations, which are subsequently filtered through the Metropolis–Hastings acceptance rule. Even when the circuit approximates the ideal dynamics imperfectly, the acceptance step ensures that the Markov chain converges to the correct stationary distribution. Consequently, circuit depth primarily affects sampling efficiency and mixing behavior rather than the correctness of the algorithm itself.

The resulting circuit (**Figure 9**) consists of alternating layers implementing the local-field and pairwise-interaction terms, providing an experimentally feasible approximation of the surrogate dynamics. Starting from an initial product configuration, the quantum device applies these rotations for a small number of Trotter steps and then performs measurements in the computational basis, producing candidate configurations that approximate samples from the target distribution $|\psi_\theta(x)|^2$. The efficiency of this quantum-assisted sampling procedure is therefore determined by how rapidly the induced Markov chain explores configuration space.

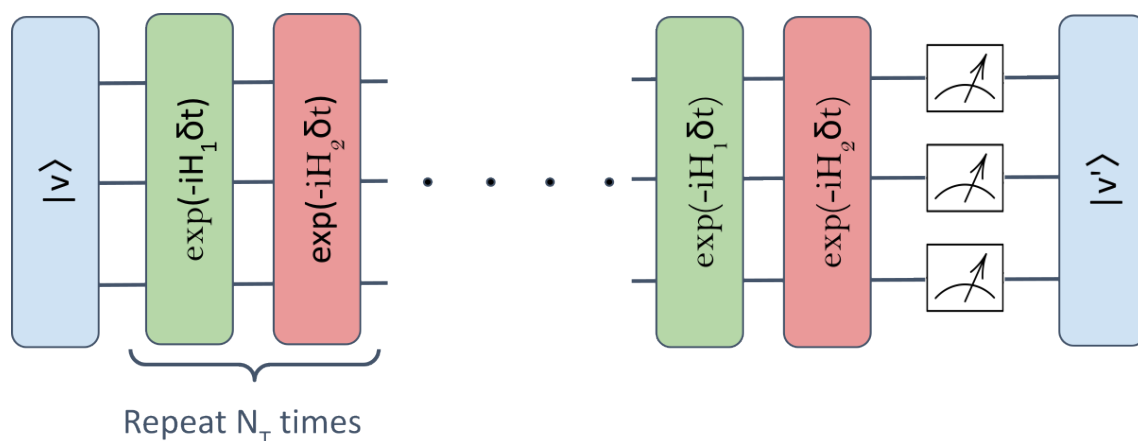


Figure 9 • Trotterized quantum circuit implementing the quantum-enhanced proposal distribution. The proposal transition probability is defined as $P_{\text{prop}}(\vec{v}^{(i+1)} | \vec{v}^{(i)}) = |\langle \vec{v}^{(i+1)} | U(\tau, \gamma) | \vec{v}^{(i)} \rangle|^2$, where the unitary evolution operator is $U(\tau, \gamma) = e^{-i\tau[\gamma h_1 + (1-\gamma)h_2]}$. The surrogate Hamiltonian consists of an Ising term $h_1 = \sum_i \ell_i(\vec{X})\sigma_z^{(i)} + \sum_{i,j} J_{ij}(\vec{X})\sigma_z^{(i)}\sigma_z^{(j)}$, whose Gibbs state encodes the approximate probability distribution of the neural-network quantum state, and a transverse-field mixer $h_2 = \sum_i \sigma_x^{(i)}$, which induces transitions between configurations. The evolution is implemented via a first-order Trotter decomposition, alternating applications of $e^{-i h_1 \delta t}$ and $e^{-i h_2 \delta t}$ for N_T Trotter steps. Measurements in the computational basis yield the proposed configuration $\vec{v}^{(i+1)}$. Reproduced from [50].

To evaluate the efficiency of this quantum-assisted sampler, the spectral gap of the corresponding transition operator is analyzed, as shown in the numerical experiments of **Figure 10**. The quantum sampler maintains a spectral gap that decays with system size at a rate much slower than the classical update MCMC techniques. This enhanced mixing time enables the Q-VMC algorithm to avoid long autocorrelation times, thereby producing statistically independent samples. As a direct consequence, gradient estimates and energy evaluations within the VMC loop exhibit significantly reduced variance, resulting in more accurate and stable convergence.

Despite the improved performance, the algorithm preserves polynomial scaling of computational resources—including circuit width, circuit depth, and the number of measurements required for gradient estimation—ensuring that the overall optimization remains efficiently implementable. The circuit width grows linearly with the number of visible units, while the depth is determined primarily by the chosen Trotter step number m . Importantly, the Trotterization does not need to accurately approximate the propagator. The sampler succeeds so long as the induced dynamics provide sufficient mixing. The number of samples required depends on the desired accuracy of the gradient estimation. Due to the low correlation among the samples generated from this method, the estimation takes significantly fewer samples compared to classical methods.

The Q-VMC algorithm is benchmarked on representative molecular Hamiltonians and spin models to assess its practical performance. **Figure 11** illustrates ground-state learning for the Li–H molecule as a function of bond stretching and for the H₂O molecule under angular distortion. In both cases, the Q-VMC approach reproduces highly accurate energies when compared to Complete Active Space Configuration Interaction (CASSCI) computations, across a broad range of geometries. These benchmarks illustrate how the quantum-assisted sampler maintains stable convergence across the same parameter ranges where standard classical NQS training slows as the geometry moves away from equilibrium.

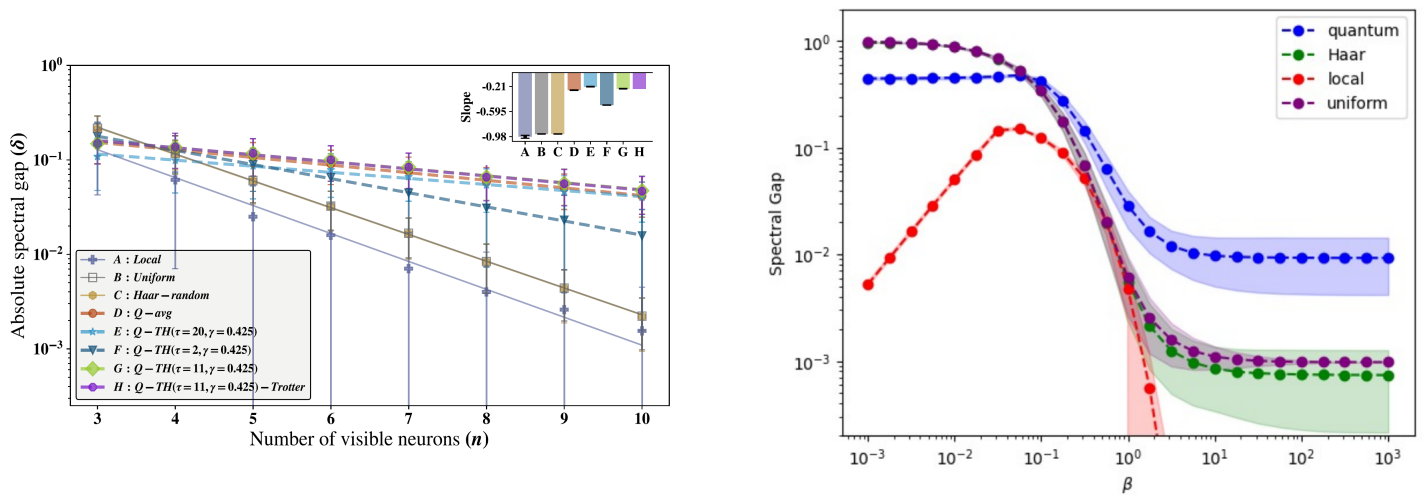


Figure 10 • (Left) Absolute spectral gap $\delta = \lambda_0 - \lambda_1$ of the transition matrix $T(\vec{v}^{(i+1)} | \vec{v}^{(i)})$ as a function of the number of visible neurons (n), for different proposal distributions labeled A–H. Classical proposals include (A) local single-spin-flip updates, (B) uniform random proposals, and (C) Haar-random proposals. Quantum-assisted proposals include (D) a quantum-averaged proposal, and (E–G) time-homogeneous (TH) quantum proposals with different evolution times τ at fixed mixing parameter γ . Case (H) corresponds to the same time-homogeneous quantum proposal implemented using a first-order Trotterized circuit. Inset shows the fitted decay slopes of δ versus (n) for each proposal. Quantum proposals (D–H) exhibit a markedly slower decay of the spectral gap compared to classical proposals (A–C), indicating improved mixing time and faster convergence to the stationary distribution. **(Right)** Spectral gap of the transition matrix as a function of inverse temperature β for different proposal distributions. Shaded regions indicate statistical uncertainty from averaging over multiple instances. The quantum proposal maintains a substantially larger spectral gap over a broad range of β , particularly in the low-temperature regime, indicating improved mixing properties compared to classical proposals. Reproduced from [50].

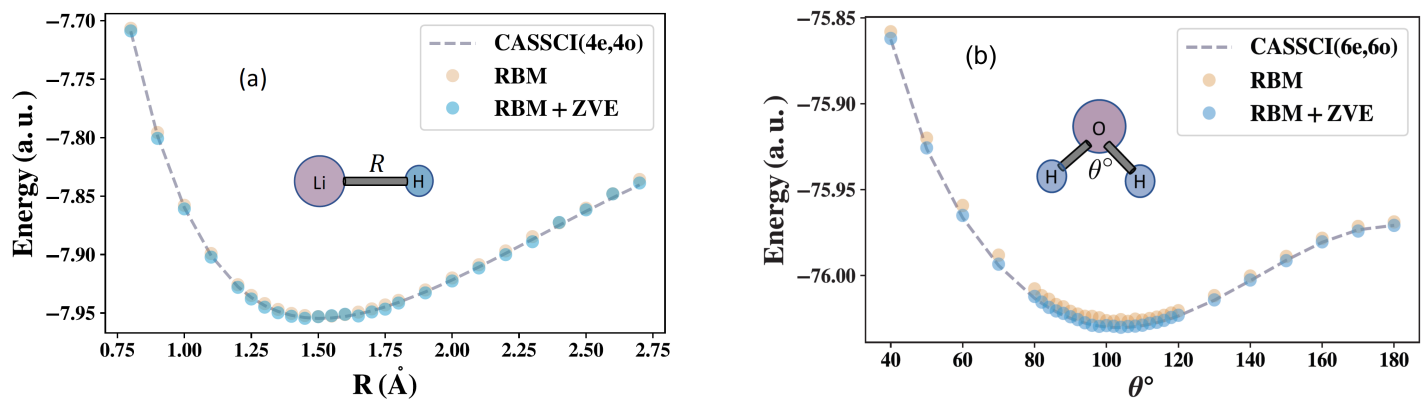


Figure 11 • Benchmarking the algorithm for ground state learning in (a) LiH molecule as a function of distortion in bond length between the Li and H atom; (b) H₂O molecule as a function of the distortion in bond angle between O and the two H atoms. For computations without ZVE (orange), the minimum energy value from the last few data points in the training protocol is used. For RBM + ZVE (blue), the y-intercept from the Zero variance extrapolation scheme is used. Both methods, especially RBM + ZVE, show good agreement with the exact CASSCI results (dashed gray line) across different bond lengths and bond angles. Reproduced from [50]. CASSCI: Complete Active Space Configuration Interaction; RBM: Restricted Boltzmann Machine; ZVE: Zero Variance Extrapolation.

The generality of the method allows it to be applied to larger and more complex NQS architectures, including deep or convolutional networks used to represent extended materials and strongly correlated quantum phases. The surrogate Hamiltonian construction accommodates a wide variety of NQS connectivity patterns, making the approach applicable beyond simple RBM-like architectures. In the high-entanglement regime, the improved mixing time manifests as faster, more stable convergence of the variational parameters, as well as more accurate Monte Carlo estimates of the energy and its gradients, which directly influence the optimization trajectory.

More broadly, the Q-VMC framework provides a blueprint for embedding quantum hardware directly into the optimization loop of NQS-based many-body solvers rather than using quantum circuits solely as variational ansätze or feature maps. This perspective suggests several avenues for future development, including quantum-enhanced training of excited states via orthogonality constraints [45, 64], quantum-assisted preparation of nonequilibrium states for open-system simulations [65–67], and NQS-based generative sampling for molecular and materials design [18, 46]. As quantum hardware continues to improve, hybrid training algorithms of this kind may offer a scalable route to

simulating complex physico-chemical systems beyond the reach of conventional classical or purely variational quantum techniques.

Assumptions and practical limitations: The main practical limitation of this framework arises from the construction of the surrogate distribution $\phi(v)$, whose parameters are obtained by fitting to samples drawn from the target distribution $\rho_{vv}(X)$. The accuracy of the surrogate, therefore, depends on how well the fitting set represents the underlying distribution. In practice, this issue can be mitigated by combining high-probability configurations with randomly selected samples to improve coverage of both dominant and background regions of the distribution, although certain target distributions may still be challenging to approximate accurately. A second consideration is the dependence of the fitting complexity on the truncation order k in the surrogate expansion of Theorem II.1. As k increases, the number of fitting parameters grows as $\mathcal{O}(N^k)$, increasing the computational cost of the fitting stage and potentially leading to over-parameterization and reduced numerical stability for large k .

These limitations primarily affect the surrogate-construction stage rather than the quantum proposal mechanism itself. Once the surrogate distribution is defined, the Metropolis–Hastings sampling procedure remains valid even if the quantum circuit implements only an approximate proposal distribution. In such cases, circuit noise or approximation errors simply reduce the quality of the proposal and may degrade performance toward that of a less-informed classical proposal, without compromising the correctness of the sampling protocol. Overall, the approach is most effective when a low-order surrogate provides a reasonable approximation to the target distribution. In this regime, the protocol maintains favorable polynomial scaling while enabling quantum-assisted sampling for training neural-network quantum states.

5. Distributed quantum learning

As quantum computing systems scale and become increasingly networked, several paradigms have emerged for performing machine learning tasks in distributed quantum environments. Broadly speaking, these approaches aim to leverage multiple quantum processors, classical nodes, or hybrid infrastructures to collaboratively train models or process quantum data without relying on a single centralized device.

One line of research focuses on distributed variational quantum algorithms, in which the parameters of a variational quantum circuit are optimized across multiple quantum processors or classical workers. Such approaches are motivated by the limited qubit counts and connectivity of near-term devices, and they enable larger models to be trained by decomposing quantum circuits or optimization tasks across multiple nodes [10, 22]. Related work has explored distributed optimization strategies, including parallel parameter updates and hybrid classical–quantum training loops designed to exploit high-performance computing infrastructure.

Another emerging direction is quantum data distribution, in which quantum datasets or measurement outcomes are processed collaboratively across geographically separated nodes. In the context of future quantum networks and the quantum internet, such approaches may enable distributed quantum sensing, secure data processing, and collaborative model training across multiple quantum devices.

More recently, ideas from classical distributed machine learning have begun to influence quantum machine learning frameworks. In particular, federated learning provides a natural paradigm for collaborative model training in which multiple clients train local models on private data and share only model updates with a central aggregator. Extending this idea to quantum settings has given rise to federated quantum learning (FQL), which aims to enable distributed training of quantum or hybrid quantum–classical models while preserving data locality and privacy. In the following subsection, we discuss this emerging paradigm in greater detail.

5.1. Quantum federated learning

Quantum federated learning (QFL) is an emerging paradigm that integrates the principles of federated learning with quantum information processing to enable collaborative model training without centralized data sharing. In classical federated learning, multiple clients train local models on their private data and exchange only model updates with a central server, preserving data privacy while avoiding the transmission of raw datasets. QFL extends this concept to quantum settings, where clients may possess quantum data, quantum models (such as variational quantum circuits), or hybrid quantum–classical architectures. By allowing quantum nodes to train local models while sharing only limited information with a central coordinator, QFL aims to combine the expressive power of quantum machine learning with the privacy and scalability benefits of distributed learning [68].

In practice, the nature of the information exchanged between clients and the central server varies across different implementations of federated quantum learning. In many existing protocols, clients locally optimize the parameters of a variational quantum circuit or hybrid quantum–classical model and periodically transmit updated model parameters to a central aggregator, following an approach analogous to federated averaging in classical machine learning. Other approaches may exchange gradient information computed from local quantum measurements, or share expectation-value statistics derived from quantum circuits that are used to update a global objective function. The specific communication strategy, therefore, determines both the computational workflow and the communication overhead of the federated system.

The privacy guarantees provided by QFL depend on the information exchanged and the assumptions made about the system architecture. In most current implementations, privacy is achieved primarily through data locality: clients retain their raw datasets locally while sharing only model updates or measurement-derived quantities. Additional security mechanisms, such as secure aggregation protocols, differential privacy, and cryptographic communication schemes, can be incorporated to further reduce the risk of information leakage during model aggregation. The level of privacy protection, therefore, depends on the adversarial model assumed and the design of the communication protocol.

Despite its promise, QFL introduces uniquely quantum challenges and opportunities. These include the presence of quantum noise, limited qubit counts, heterogeneous quantum hardware across clients, and the difficulty of coordinating distributed training across devices with varying error rates and circuit capabilities. Advanced QFL frameworks, therefore, explore techniques such as layer-wise or parameter-wise aggregation of variational quantum circuits, quantum-aware optimization

strategies, and noise-resilient protocols designed to handle non-IID datasets and device-dependent errors. As quantum hardware matures, QFL is expected to play an important role in scalable and privacy-aware quantum intelligence, enabling collaborative learning across quantum networks, cloud-based quantum processors, and future quantum internet infrastructures.

Figure 12 illustrates a typical QFL workflow. A global quantum model is first initialized on a central server and broadcast to participating clients. Each client then performs local training using its private dataset and updates the parameters of its local quantum model. Instead of sharing raw data, clients transmit only model updates to the central aggregator. The server combines these updates using federated aggregation strategies to obtain an improved global model, which is subsequently redistributed to clients for further training. This iterative procedure enables collaborative optimization of quantum models while preserving data locality and reducing privacy risks associated with centralized data collection [69].

5.2. Applications of federated quantum machine learning

Federated quantum learning has recently been explored in several application domains where datasets are naturally distributed across institutions and cannot be centrally aggregated. Prominent examples include healthcare analytics, medical imaging, and drug discovery, where privacy regulations and institutional constraints limit direct data sharing. In such settings, QFL enables collaborative model training across multiple organizations while ensuring that sensitive data remains locally stored.

Several studies have investigated the use of hybrid quantum-classical architectures in federated environments, including quantum convolutional neural networks, quantum convolutional neural networks, and variational quantum circuits. These models can be trained collaboratively across distributed datasets while exchanging only model updates rather than raw data. Such approaches have demonstrated promising performance in classification tasks involving medical imaging datasets such as MedNIST, PneumoniaMNIST, and CT kidney disease datasets [69–71]. These early results suggest that federated quantum learning may improve model generalization across heterogeneous data sources while maintaining strong privacy guarantees.

Figure 13 illustrates the behavior of federated quantum models under non-IID data distributions, a common scenario in distributed healthcare datasets where each institution may observe different subsets of patient populations or imaging modalities. Experimental results demonstrate that federated quantum models can successfully learn across heterogeneous data partitions while maintaining competitive accuracy.

Recent work has also explored federated training of variational quantum circuits for medical diagnosis tasks, including prediction of coronary heart disease, diabetes detection, and cancer classification across datasets distributed among multiple hospitals [71]. **Figure 14** illustrates an example of such a framework, where local quantum models trained at different institutions contribute updates to a shared global model. These studies show that quantum-aware optimization strategies, such as quantum natural gradient methods, can improve convergence behavior and reduce the number of communication rounds required during federated training.

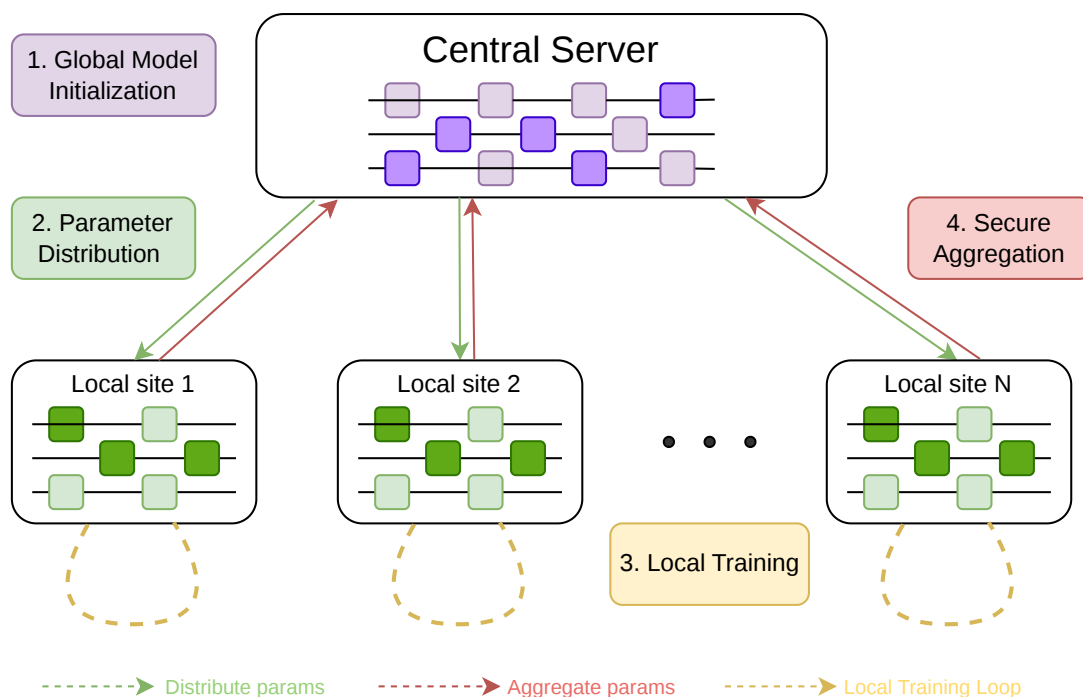


Figure 12 • Illustration of a quantum federated learning (QFL) architecture for distributed training of parameterized quantum models across multiple local sites. (1) A central server initializes a global quantum model (e.g., a parameterized circuit) and (2) distributes its parameters to participating clients. (3) Each local site performs independent training using private data through a local optimization loop, updating circuit parameters without sharing raw data. (4) The updated parameters are returned to the server and combined via secure aggregation to produce an improved global model. This iterative process enables collaborative learning while preserving data privacy and leveraging the expressivity of quantum models across heterogeneous data sources.

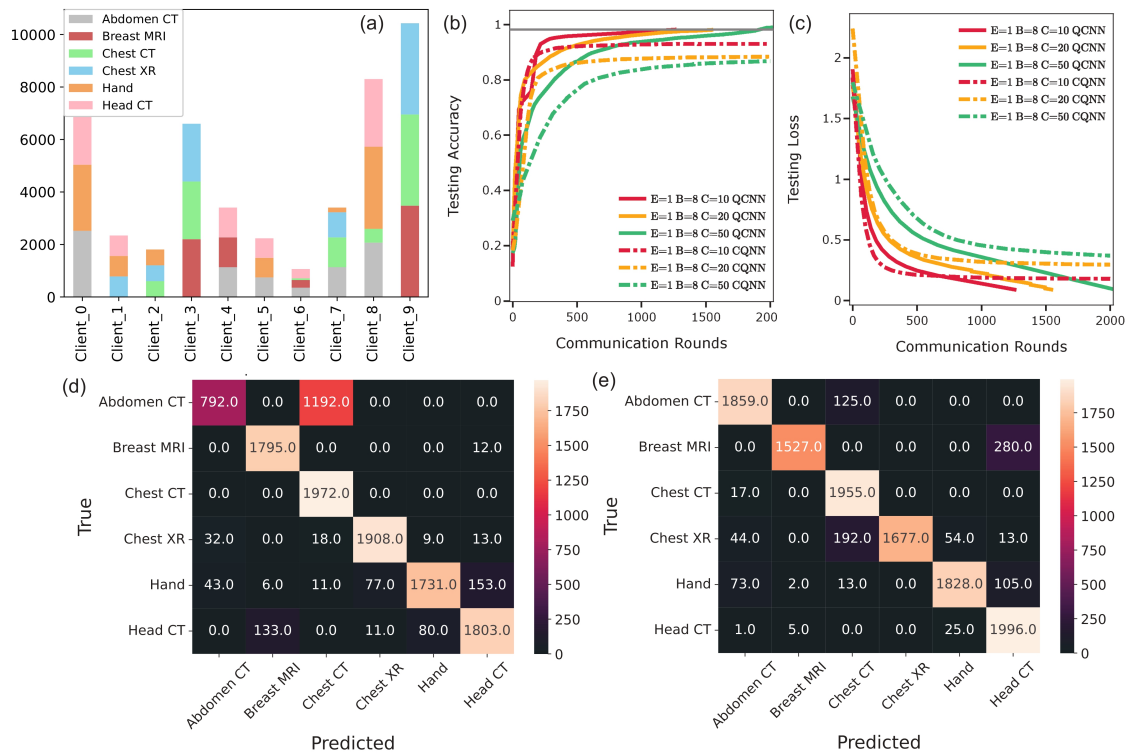


Figure 13 • A performance comparison between federated quanvolutional QCNN and classical-quantum CQNN on the non-IID2 partition of the MedNIST dataset is illustrated. (a) The non-IID2 distribution shows that each of the 10 clients receives random training samples from at least three different classes. (b) The plot depicts testing accuracy versus communication rounds for the Non-IID2 MedNIST dataset with 10, 20, and 50 clients. (c) The validation loss curves indicate that the CQNN model converges faster and more smoothly. (d,e) Confusion matrix for the testing set after training for 100 rounds, and finally classified the relevant classes. Reproduced from [70]. QCNN: Quantum Convolutional Neural Network; CQNN: Classical-Quantum Neural Network; CT: Computed Tomography; MRI: Magnetic Resonance Imaging; XR: X-ray.

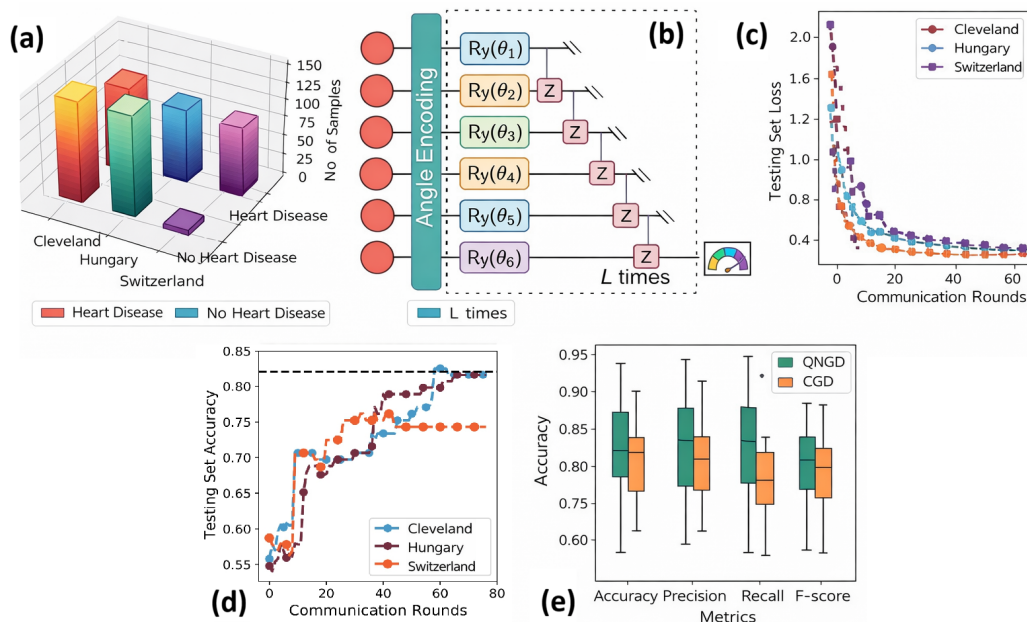


Figure 14 • (a) A three-dimensional bar chart illustrating binary classification outcomes for Heart Disease and No Heart Disease categories, together with the corresponding sample contributions from the Cleveland, Hungary, and Switzerland hospital datasets. (b) A 13-qubit variational quantum circuit employing angle encoding via Ry rotations and a linear entanglement pattern. (c,d) Testing loss and testing accuracy curves demonstrating the performance of locally trained models from the three hospitals on the Heart Disease (HD) dataset. The black-dotted horizontal line denotes the target accuracy achieved by training on the full dataset without federated learning. (e) Evaluation of the performance and robustness of the FedVQC model across different datasets using quantum natural gradient descent (QNGD) and classical gradient descent (CGD) optimizers. Model performance is assessed using standard classification metrics, including accuracy, precision, recall, and F1-score, on the test set, with visual comparisons of testing accuracy and loss for both QNGD- and CGD-optimized FedVQC models. Reproduced with permission from [71].

Overall, these studies highlight the potential of federated quantum learning as a framework for privacy-preserving collaborative AI. By combining distributed machine learning with quantum or hybrid quantum–classical models, QFL enables institutions to jointly improve predictive models while maintaining strict data governance. Although current demonstrations remain largely proof-of-concept, federated quantum learning may play an important role in future distributed quantum computing ecosystems, particularly in domains where privacy, data locality, and cross-institution collaboration are essential.

6. Applications of quantum machine learning to complex systems

Quantum machine learning (QML) has increasingly transitioned from a primarily theoretical research direction to a practical framework for addressing challenging scientific and engineering problems. In particular, the ability of quantum and hybrid quantum–classical models to represent highly correlated, high-dimensional systems has motivated their application to a variety of complex physical, chemical, and biological domains. By combining quantum computational primitives with machine learning-based representations and optimization strategies, QML provides new tools for exploring large configuration spaces, learning effective models of quantum systems, and accelerating simulation workflows [6].

These capabilities have led to growing interest in applying QML across a broad spectrum of complex systems, including quantum chemistry and materials science, condensed-matter physics, and non-equilibrium quantum dynamics. Beyond fundamental physics and chemistry, emerging applications are also being explored in areas such as drug discovery [72], healthcare and biomedical analytics [69, 70], life sciences [73, 74], semiconductor design [75], and financial portfolio optimization [76]. In the following subsections we highlight several representative application areas where QML techniques are beginning to provide new capabilities for modeling and analyzing complex systems.

6.1. Quantum chemistry and molecular simulation

One of the earliest and most widely explored application areas of quantum machine learning is the simulation of molecular and electronic structure problems. Accurately modeling the ground and excited states of molecules remains a central challenge in quantum chemistry due to the exponential growth of Hilbert space with system size and the presence of strong electronic correlations. Quantum machine learning approaches have been proposed to assist both classical and quantum simulation workflows by learning compact representations of molecular wavefunctions, reduced density matrices, and energy landscapes [11, 12, 77, 78].

In hybrid quantum–classical frameworks, parameterized quantum circuits or neural-network quantum states can be trained to approximate molecular wavefunctions or electronic energies. These models have been explored for tasks such as variational electronic structure calculations, state preparation, and observable estimation in molecular systems. More broadly, machine learning techniques are increasingly used to accelerate electronic structure calculations, identify effective Hamiltonians, and guide quantum simulations of chemical processes on near-term quantum hardware [11, 79]. Such approaches are particularly

promising for systems with strong correlations or complex geometries, where conventional computational chemistry methods become increasingly expensive.

6.2. Condensed-matter systems and quantum materials

Quantum machine learning has also found applications in condensed-matter physics and materials discovery, where many-body interactions and collective phenomena lead to rich emergent behavior. Machine learning models have been used to identify phases of matter, detect quantum phase transitions, and characterize topological order in interacting systems. In these contexts, neural-network-based wavefunction representations and variational quantum circuits provide flexible ansätze capable of capturing complex correlations in quantum many-body states [11, 77, 78].

Beyond phase identification, QML techniques have been applied to the study of quantum materials and strongly correlated lattice models. Machine learning models can assist in exploring large parameter spaces associated with candidate materials, predicting electronic properties, and identifying effective low-energy models that capture experimentally relevant phenomena. These methods offer new tools for navigating the high-dimensional design spaces that arise in materials discovery and quantum many-body simulation.

6.3. Non-equilibrium quantum dynamics

While many early QML studies focused on ground-state properties, a growing body of work has extended machine learning approaches to the simulation of real-time and far-from-equilibrium quantum dynamics. In the neural-network quantum state framework, time evolution can be formulated using the time-dependent variational principle (TDVP), which projects the Schrödinger equation onto a parameterized manifold of wavefunctions. This approach enables approximate real-time evolution of interacting many-body systems by solving differential equations for the network parameters [51, 80].

Complementary progress has emerged in hybrid quantum algorithms for dynamical simulation. Variational quantum simulation (VQS) recasts both real-time and imaginary-time evolution as optimization problems over parameterized quantum circuits, allowing approximate propagation using shallow circuits compatible with noisy intermediate-scale quantum devices [65, 66].

Open quantum systems present additional challenges because their dynamics are governed by non-unitary evolution, typically described by Lindblad master equations. Recent work has extended neural-network and variational-circuit approaches to represent mixed states and dissipative dynamics, enabling approximate simulation of steady states and driven many-body systems [42, 67, 81–83]. These developments broaden the scope of quantum machine learning beyond equilibrium physics, providing new tools for studying driven, dissipative, and strongly interacting quantum systems.

6.4. Drug discovery

In contemporary drug discovery, accurately evaluating the absorption, distribution, metabolism, and excretion (ADME)

profiles, alongside toxicity (Tox) properties of novel chemical entities, remains a significant bottleneck [84]. These assessments are inherently time-intensive, costly, and present major challenges for pharmaceutical research and development. Recent advances in computational sciences, including artificial intelligence (AI), big data analytics, and cloud computing, have begun to transform this landscape by enabling predictive models for ADME-Tox properties [85]. Emerging at the forefront of this transformation is the integration of quantum computing with machine learning, which has shown promise across chemistry, biomedicine, and engineering. Quantum computational approaches promise to accelerate high-throughput screening of molecular libraries, enabling the evaluation of billions of compounds while substantially reducing both time and cost associated with drug development. Here, we present a hybrid quantum–classical machine learning framework designed to predict ADME-Tox properties of chemical compounds [72]. The framework integrates a classical support vector classifier with a kernel-based quantum classifier, leveraging the efficiency of quantum kernel methods. Specifically, chemical structures encoded via Simplified Molecular Input Line Entry System (SMILES) strings are transformed into a quantum feature space using string kernels, followed by classification using a quantum-enhanced support vector machine [86]. To assess the performance and practicality of this approach, we conducted extensive simulations on large-scale chemical datasets. Our results demonstrate that the proposed quantum machine learning framework can effectively capture complex molecular features and offers a promising avenue for accelerating ADME-Tox predictions in drug discovery pipelines. Leveraging the potential computational benefits and noise resilience of quantum support vector classifiers, we introduce a quantum machine learning (QML) framework employing QSVC to predict ADME-Tox properties of chemical compounds for drug discovery applications, as depicted in **Figure 15**.

The framework consists of: (1) ADME-Tox datasets along with molecular structures and SMILES representations, (2) feature preprocessing and dimensionality reduction using PCA, (3) a quantum classifier implemented on both quantum simulators and hardware to predict ADME-Tox properties using four datasets [87], and (4) evaluation of model performance using classification metrics. In **Figure 15**, we present the overall workflow of the proposed framework. To comprehensively investigate the diversity and complexity of ADME-Tox properties, we employed four benchmark chemical compound datasets [87]: Human Intestinal Absorption (HIA), CYP2D6 Substrate, Drug-Induced Liver Injury (DILI), and Carcinogenicity (CARC). These datasets enable accurate feature learning based on the structural information of drug candidates. All datasets were obtained from the Therapeutic Data Commons (TDC), which provides curated resources for AI/ML tasks in multiple therapeutic modalities, including small molecules, macromolecules, and cell and gene therapies to support effective drug development and improved patient safety throughout the lifecycle of the medicine. The support vector classifier (SVC) is a well-known classical machine learning algorithm, also called the ‘kernel’ method, for classification and regression. Suppose that given N training data points (x_i) and associated labels $y_i \in \{-1, 1\}$, which are spelled in dimensions D , where $1 \leq i \leq N$. The objective is to find a hyperplane that maximizes the margin between the classes in space. In SVC, it computes the inner product between pairs of datapoints $\langle x_i | x_j \rangle \in \mathbb{R}$. If the data points cannot be separated linearly, then transform

the input data points to a high-dimensional feature space (φ) via a mapping function and determine a hyperplane with maximum margin. The technique ‘kernel trick’ in SVC enables handling non-linear datapoints, which relies on the inner product between pairs of mapping kernel functions such that $K(x_i, x_j) = \varphi(x_i) \cdot \varphi(x_j)$. Thus, the optimization objective function of SVC with the kernel trick is given as:

$$\begin{aligned} \max f(c_1, \dots, c_N) &= \sum_{i=1}^N c_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y_i c_i (\varphi(x_i) \cdot \varphi(x_j)) y_j c_j \\ &= \sum_{i=1}^N c_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y_i c_i K(x_i, x_j) y_j c_j \end{aligned} \quad (11)$$

subject to $\sum_{i=1}^N c_i y_i = 0$, and $0 \leq c_i \leq 1/2N\lambda$, where λ controls the hard margin classifier. Although there exist various kernel functions, such as linear, polynomial, Gaussian (RBF), and sigmoid, to solve different problems. Kernel-based learning methods have been successfully applied to problems involving high-dimensional input spaces. The quantum support vector classifier (QSVC) [88] is an extension of the classical support vector classifier (SVC) that leverages a quantum feature space to enhance computational power. The quantum feature space $|\varphi(x_i)\rangle \langle \varphi(x_j)|$ is a high-dimensional complex-valued Hilbert space, which allows for the exploitation of quantum computing capabilities. An architecture of QSVC is shown in **Figure 16**. Here, a quantum kernel function is employed to map the data into a quantum feature space represented as

$$K(x_i, x_j) = |\langle \varphi(x_i) | \varphi(x_j) \rangle|^2 \quad (12)$$

where $\varphi(x_i)$ denotes the first step is to encode classical points $x \in \mathbb{R}^n$ into a quantum data using a n -qubit parameterized quantum circuit (PQC) $|\varphi(x)\rangle = U(x)|0^n\rangle$, where U is a unitary operation. We used the ZZFeaturemap with a linear entanglement layer (depth = 2). Although we can vary the depth of the feature map circuit to introduce more entanglement and repeat ($\times l$) the encoding circuit. The number of qubits depends on the number of features outlined in our dataset. The quantum kernel is estimated by evolving a reference state $|0\rangle^n$ as

$$K(x_i, x_j) = |\langle 0^n | U^\dagger(x_i) U(x_j) | 0^n \rangle|^2 \quad (13)$$

The classification performance of QSVC across ADME and toxicity datasets is summarized in **Table 2**. The quantum circuit with linear entanglement for 8-dimensional classical input features is given in **Figure 17**, where H denotes the Hadamard gate to create superposition, and $P(\theta)$ is a phase gate. A unitary operation (U) is applied to quantum registers initialized in the $|0\rangle$ state, and then the inverse operation $U^\dagger(x)$ is applied. Finally, a measurement operator is applied to estimate the quantity of interest (i.e., the target variable). To showcase the potential advantage of a quantum machine learning algorithm, the proposed framework conducted experiments in two phases: In the initial phase, the experiments were executed in a noiseless quantum environment with the python-based Qiskit’s statevector simulator (SV), and qasm simulator (QA), which means that the simulations assumed perfect conditions without any form of noise or errors [89]. The QSVC model was evaluated on balanced HIA, CYP2D6, DILI, and CARC datasets, showing that performance consistently improves as feature dimensionality increases from 2 to 16, while remaining compatible with current quantum hardware limits. QSVC achieved strong predictive capability for HIA and CARC ($AUC \approx$

0.93–0.95) and stable performance for DILI (AUC $\approx$ 0.85–0.87), with robustness even under noise. CYP2D6 remained the most challenging task, performing moderately (AUC $\approx$ 0.77–0.79) and below a fully featured classical SVC, indicating that some molecular properties may require higher-dimensional embeddings. Compared with classical models, QSVC often matched or exceeded performance using far fewer features, remained consistent on

real IBM hardware, and demonstrated reliable generalization as shown in **Figure 17**. Overall, the results highlight that QSVC effectively captures complex molecular relationships, delivers competitive ADME-Tox prediction with reduced feature requirements, and shows promising signs of emerging quantum advantage as hardware capability grows.

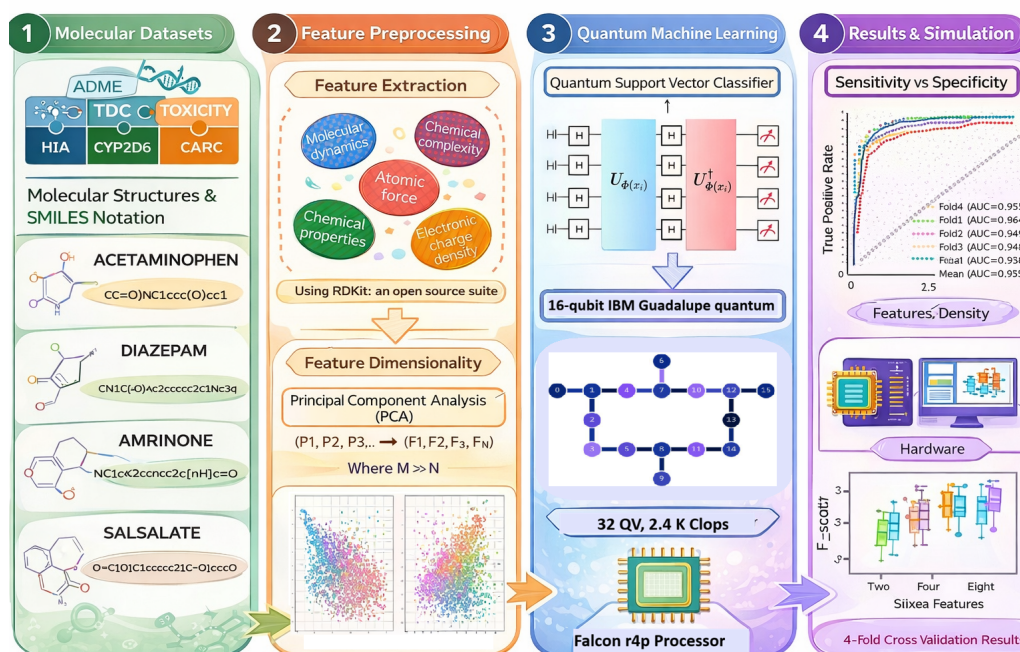


Figure 15 • An overview of a quantum machine learning framework for predicting ADME and toxicity properties. (1) ADME-Tox datasets and their molecular structures and SMILES notations. (2) Feature processing and feature dimension reduction using PCA. (3) Quantum classifier for predicting ADME-Tox properties on quantum simulator and hardware. (4) Classification metrics. Reproduced with permission from [72]. ADME: Absorption, Distribution, Metabolism, and Excretion; TDC: Therapeutic Data Commons; HIA: Human Intestinal Absorption; CARC: Carcinogenicity; AUC: Area Under the Curve; PCA: Principal Component Analysis; SMILES: Simplified Molecular Input Line Entry System.

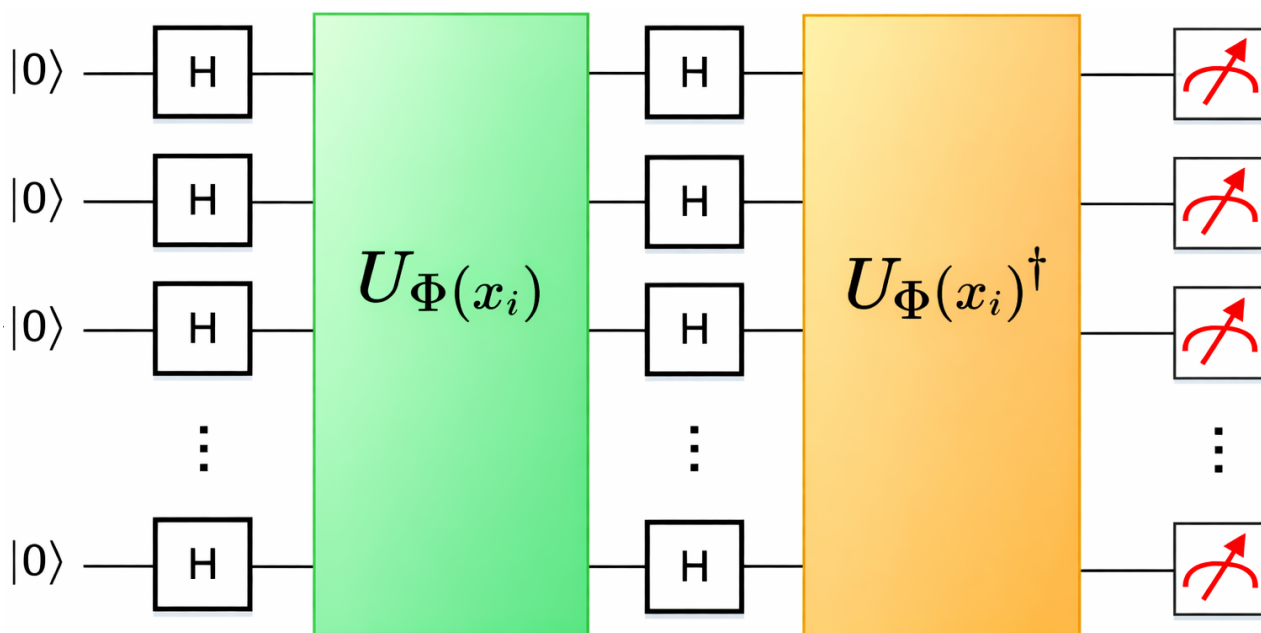


Figure 16 • A generalized quantum circuit architecture of quantum support vector classifier. It consists of the encoding circuit, followed by processing (feature map embedding and its inverse) and measurement at the end. Reproduced with permission from [72].

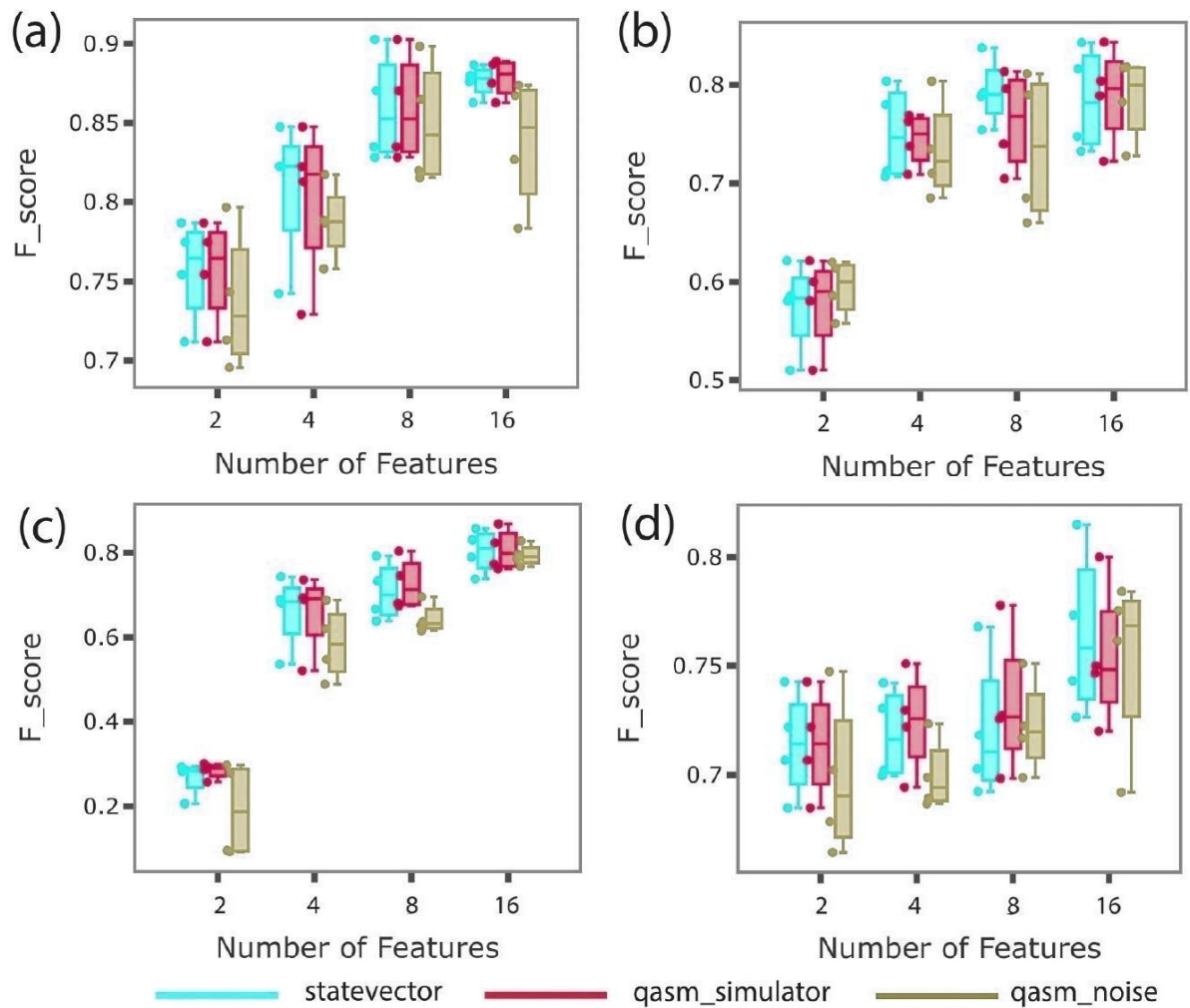


Figure 17 • Graphical representation of boxplots for F-score performance metric of quantum simulators (statevector in orange; qasm simulator in blue; qasm with noise in purple). Here, we applied 4-fold cross-validation across all simulations and presented the F-scores for all datasets (CARC, DILI, HIA, and CYP2D6) with different numbers of features for all quantum simulators in (a–d), respectively. In boxplots, the median value is denoted by the horizontal line, and the box defines upper and lower quantiles. The data points are the statistical outliers. Reproduced with permission from [72]. CARC: Carcinogenicity; DILI: Drug-Induced Liver Injury; HIA: Human Intestinal Absorption.

Table 2 • Classification performance metrics of QSVC considering 16 features on QASM noise model. Reproduced with permission from [72].

Metrics	ADME dataset: HIA				ADME dataset: CYP2D6			
	Fold 1	Fold 2	Fold 3	Fold 4	Fold 1	Fold 2	Fold 3	Fold 4
Accuracy	0.863	0.881	0.881	0.863	0.706	0.664	0.746	0.735
Precision	0.703	0.750	0.774	0.777	0.842	0.730	0.847	0.752
Recall	0.918	0.785	0.889	0.792	0.694	0.657	0.729	0.800
F-score	0.796	0.767	0.827	0.785	0.761	0.691	0.784	0.775
Metrics	Toxicity dataset: DILI				Toxicity dataset: CARC			
	Fold 1	Fold 2	Fold 3	Fold 4	Fold 1	Fold 2	Fold 3	Fold 4
Accuracy	0.757	0.841	0.766	0.849	0.821	0.785	0.845	0.747
Precision	0.796	0.862	0.775	0.872	0.907	0.895	0.900	0.844
Recall	0.741	0.814	0.789	0.842	0.830	0.767	0.849	0.730
F-score	0.727	0.818	0.782	0.817	0.867	0.826	0.873	0.783

QSVC: Quantum Support Vector Classifier; QASM: Quantum Assembly Language; ADME: Absorption, Distribution, Metabolism, and Excretion; HIA: Human Intestinal Absorption; DILI Drug-Induced Liver Injury; CARC: Carcinogenicity.

6.5. Cancer biology and biomarker discovery

Quantum machine learning offers novel opportunities to resolve, accelerate, and refine computational biology challenges. In [73], our study aims to investigate the performance of multi-omic data integration and high-dimensional feature encoding for cancer prediction (primary tumor or normal solid tissue) and to assess feature importance using quantum–classical machine learning. Moreover, in [74], we focus on advancing QML for biomedical applications, particularly in multi-omic data integration for lung cancer subtype classification. Traditional machine learning approaches often struggle with the high dimensionality and heterogeneity of omics data. To address this, we developed a Multi-omic Quantum Machine Learning Lung Subtype Classification (MQML-LungSC) framework that leverages Quantum Neural Networks (QNNs) for enhanced classification performance. By encoding multi-omic features, including DNA methylation, RNA-seq, and miRNA-seq, our approach identifies key molecular markers distinguishing lung adenocarcinoma (LUAD) from lung squamous cell carcinoma (LUSC). Our results demonstrate that QNNs, particularly with 256 encoded features, outperform classical machine learning models such as Logistic Regression (LR), Multi-Layer Perceptron (MLP), and support vector machines (SVM). The QNN256 model achieved training and test accuracies of 0.95 and 0.90, respectively, with AUCs of 0.99 and 0.96, respectively. Additionally, our methodology enables the identification of top-hit biomarkers using model weights, facilitating biomarker discovery for potential clinical applications. This study underscores the promise of quantum–classical hybrid models in personalized medicine and sets the stage for future advancements in QML-driven cancer diagnostics. **Figure 18** shows the workflow of multi-omics integration and classification using quantum neural networks. **Figure 18a** Schematic representation of the overall pipeline of the MQML-QNN framework. Data acquisition from GDC-TCGA, including (i) DNAm, (ii) RNA-seq, (iii) miRNA-seq, and (iv) clinical and survival attributes of patients. Data preprocessing and feature engineering using t-test p-values for each omic data type to differentiate between subtypes I and II. Feature selection process involving: (i) Four feature selection models—RF, MI, PCA, and Chi; (ii) AUC-ROC analysis; (iii) hierarchical clustering based on sample vs. feature, and pairwise feature similarity using distance metrics. Integration of multi-omic data resulted in 256 features from 915 common patients. Quantum amplitude encoding with three feature dimensions—32, 64, and 256—and the quantum circuit with a dense layer for diagnostic classification of LUAD versus LUSC lung datasets. **Figure 18b,c** Visualization of classification results through line plots, confusion matrices, and heatmaps.

6.6. Agro-climate modeling

In recent years, quantum computing has emerged as a powerful tool in agriculture, particularly for processing highly correlated, high-dimensional climate data [90]. Matrix Product State (MPS) quantum classifiers [91], a type of one-dimensional tensor network, have been shown to efficiently encode classical agricultural data, such as temperature, humidity, wind speed, solar radiation, sunshine hours, and evapotranspiration (ET_o) [92, 93], into quantum-entangled states for classification tasks. MPS constrains the extent of entanglement through its bond dimensions. It is a tensor network method in which tensors are connected in a

one-dimensional geometry. In principle, any pure quantum state can be represented by appropriately decomposing its coefficients. In the MPS formalism, a pure quantum state $|\phi\rangle$ is expressed as

$$|\phi\rangle = \sum_{\sigma_1, \sigma_2, \dots, \sigma_L} \text{Tr}[M_1^{\sigma_1} M_2^{\sigma_2} \cdots M_L^{\sigma_L}] |\sigma_1, \sigma_2, \dots, \sigma_L\rangle, \quad (14)$$

where $M_i^{\sigma_i}$ are complex square matrices, d denotes the local dimension, σ_i represents the local basis indices (i.e., $\{0, 1\}$ for qubits), and $\text{Tr}(\cdot)$ denotes the trace of the matrices. In quantum mechanics, the N independent systems can be combined by performing the tensor product operation on their respective state vectors [94]. The feature map is considered as shown in **Equation (15)**:

$$\phi_d(x) = \phi_{s_1}(x_1) \otimes \phi_{s_2}(x_2) \otimes \cdots \otimes \phi_{s_N}(x_N), \quad (15)$$

where s_j are indices running over the local dimension d such that $d = \{s_1, s_2, \dots, s_N\}$. Therefore, each state vector x_j is mapped to the full feature map $\phi(x)$ in a d -dimensional space. Before illustrating the MPS tensor network, it is crucial to encode a classical machine learning dataset into a quantum state. Consider a classical dataset $S = \{(x^d, y^d)\}_{d=1}^D$ for binary classification, where $y^d \in \{0, 1\}$ are class labels for N -dimensional input vectors such that $x^d \in \mathbb{R}^N$. The input vectors are normalized to lie in $[-\pi, \pi]$. Thus, the qubit ϕ is represented as in **Equations (16) and (17)**:

$$\phi_n^d = \cos(x_n^d) |0\rangle + \sin(x_n^d) |1\rangle, \quad (16)$$

$$\phi_n^d = \begin{bmatrix} \cos(x_1^d) \\ \sin(x_1^d) \end{bmatrix} \otimes \begin{bmatrix} \cos(x_2^d) \\ \sin(x_2^d) \end{bmatrix} \otimes \cdots \otimes \begin{bmatrix} \cos(x_N^d) \\ \sin(x_N^d) \end{bmatrix}. \quad (17)$$

The N -dimensional input vectors $x^d \in \mathbb{R}^N$ are mapped to a product state on N qubits by using the feature map as shown in **Equation (15)**. The full quantum data is represented as the tensor product $\phi^d = \bigotimes_{n=1}^N \phi_n^d$ [95]. Thus, the preparation of the quantum state is efficient as it only needs single-qubit rotations to encode each segment of the classical dataset $n = \{1, 2, \dots, N\}$ in the amplitude of a qubit, with no high cost for such encoding. Similar to a classical dataset for binary classification, a quantum dataset for binary classification is denoted as a set $S_q = \{(\phi^d, y^d)\}_{d=1}^D$, where $y^d \in \{0, 1\}$ are class labels for 2^N -dimensional input vectors such that $\phi^d \in \mathbb{C}^{2^N}$. It can be easily verified that the quantum data, as the output of a quantum circuit, is in a superposition state. To assess the quality of the actual and predicted values in the dataset, a cost function is needed. It measures the difference between actual and predicted values and is given by **Equation (18)**:

$$J_\theta = \frac{1}{D} \sum_{d=1}^D \left(M_\theta(x^d) - y^d \right)^2, \quad (18)$$

where x^d and y^d are the input and class labels respectively, M is a qubit operator, θ represents the set of parameters defining the unitaries, and D is the total number of data points. This calculates the average deviation of the model's predictions from the actual values. The goal is to minimize the cost function; i.e., it must be close to zero. In this context, historical meteorological data were preprocessed, normalized, and categorized into binary classes of ET_o (LOW, MEDIUM, HIGH) to form distinct classification tasks [96]. The MPS quantum circuit, composed of single-qubit rotations, CNOT gates, and ancilla-assisted nonlinear unitaries, was trained on 80% of the data and tested on the remaining 20%, achieving accuracies of 73–80% across samples, along with high sensitivity and specificity.

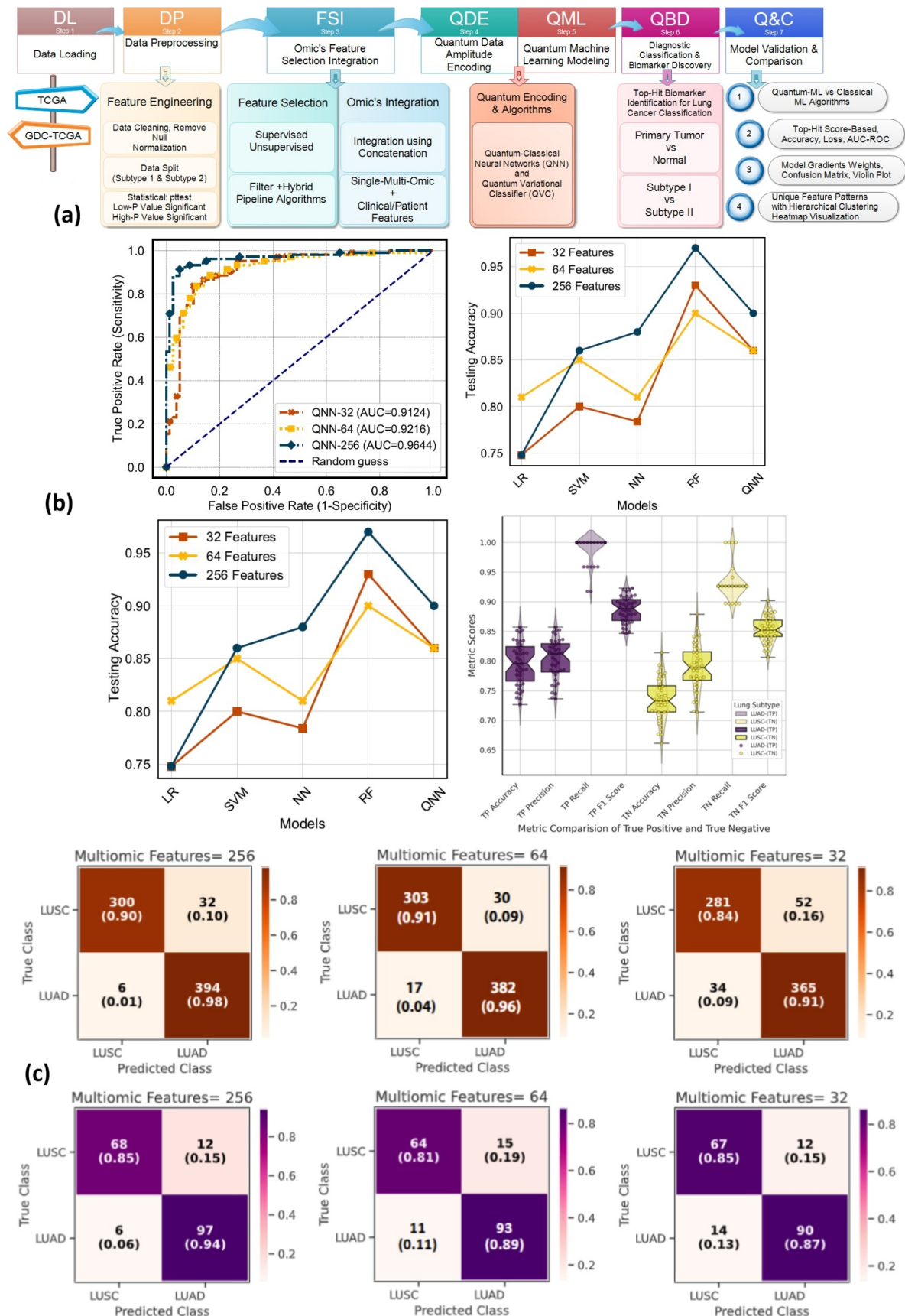


Figure 18 • Workflow of multi-omics integration and classification using a quantum neural network (MQML-QNN). (a) Overall pipeline including multi-omics data acquisition (DNA methylation, RNA-seq, miRNA-seq, and clinical data), preprocessing, feature selection (RF, MI, PCA, Chi), and integration into 256 features followed by quantum amplitude encoding and QNN-based classification of LUAD vs. LUSC. (b) Line plots showing classification performance. (c) Confusion matrices illustrating prediction accuracy. Reproduced from [74]. MQML-QNN: Multi-omic Quantum Machine Learning Quantum Neural Network; RF: Random Forest; MI: Mutual Information; PCA: Principal Component Analysis; LUAD: Lung Adenocarcinoma; LUSC: Lung Squamous Cell Carcinoma; LR: Logistic Regression; SVM: Support Vector Machine; NN: Neural Network; TN: True Negative; TP: True Positive.

In the original manuscript, the authors illustrate the overall architecture and processing steps of the proposed MPS quantum classifier applied to the Indian Meteorological Department (IMD), Pune weather dataset. Daily meteorological data for Patiala in 2014, 2015, and 2016 have been used. It consists of the following parameters: maximum and minimum air temperature ($T_{\max}$, $T_{\min}$), relative humidity (RH), wind speed (u_2), solar radiation (R_s), sunshine hours (I_s), and evapotranspiration (ET_o). Initially, the weather dataset is partitioned pairwise into three samples: Agri1 (Sample1), Agri2 (Sample2), and Agri3 (Sample3). The dataset is then normalized to ensure uniform feature scaling before applying binary classification, where the input feature vectors $X_1, X_2, \dots, X_n$ and corresponding target labels $Y_1, Y_2, \dots, Y_n$ represent low, medium, and high categories for the estimation of reference evapotranspiration (ET_o). The combined pairwise samples are further divided into training and test subsets. The training and testing datasets are mapped into a tensor network state using **Equation (15)**. Each classical input vector is encoded into a quantum state using **Equation (16)**, enabling the representation of classical weather parameters in a quantum Hilbert space. Subsequently, the tensor product of each input quantum state is taken to form complete quantum data, as defined in **Equation (17)**, which is suitable for processing within a quantum circuit. This representation corresponds to a Matrix Product State with multiple sites. The encoded quantum states are then processed by the MPS quantum classifier, which performs tensor contractions and measurements to obtain classification results. Finally, the validation process evaluates the model's performance on the test dataset, yielding classification outputs for the low, medium, and high ET_o categories. The method efficiently captures correlations among climatic variables, mitigates overfitting through adaptive bond dimensions, and requires a small number of qubits, making it suitable for near-term quantum hardware. Visualization confirmed a strong correspondence between predicted and actual ET_o values. The advantages of this approach include efficient handling of large datasets, robust learning capability, and potential scalability to other agricultural parameters. Practical applications include precision irrigation planning, climate-adaptive farming, and integration into smart agriculture systems for real-time decision-making [93]. Overall, this work demonstrates that MPS quantum classifiers provide a promising framework for leveraging quantum computing to enhance agricultural data analysis and improve resource management.

7. Conclusions

This review illustrates the ongoing research in quantum machine learning that is particularly relevant to complex systems, emphasizing approaches that address practical challenges in representation, sampling, learning dynamics, and deployment. We provide an account of how QML is evolving from a collection of conceptual proposals into a framework capable of engaging with realistic scientific and data-driven problems. We first examined quantum-enabled variational Monte Carlo strategies for training neural-network quantum states, highlighting how quantum resources can be integrated into classical learning loops to mitigate sampling bottlenecks in strongly correlated systems. We then reviewed emerging efforts to understand the learning dynamics of QML models, where tools from quantum information and many-body physics provide insight into optimization landscapes, trainability, and stability beyond conventional performance metrics.

Together, these developments underscore that scalable QML requires not only expressive models but also principled mechanisms for efficient training and diagnostic understanding. Building on this methodological foundation, we surveyed application-driven studies in domains such as drug discovery, cancer biology, and agro-climate modeling, where high-dimensional structure, limited data, and domain constraints pose stringent challenges for learning algorithms. These applications serve both as motivation and benchmarks for QML, revealing the conditions under which hybrid quantum–classical approaches may offer practical advantages. Finally, we discuss federated quantum machine learning as a pathway toward distributed and privacy-preserving learning, particularly relevant for domains such as healthcare, materials science, and large-scale scientific collaborations where data sharing is restricted. In the context of materials design, federated QML enables multiple laboratories or industrial partners to collaboratively train quantum models on locally generated simulation or experimental data, such as electronic structure calculations, spectroscopy, or synthesis outcomes, without exposing proprietary datasets. By aggregating learned quantum features rather than raw data, this approach can accelerate the discovery of functional materials, optimize design pipelines, and support secure multi-institutional innovation while respecting intellectual property and data governance constraints.

Looking ahead, the continued development of QML will depend on progress on multiple fronts, including improved quantum hardware, quantum error correction, scalable training protocols, and system-level integration with classical infrastructure. Equally important will be a deeper theoretical understanding of when and why quantum resources improve learning performance for complex systems. By consolidating insights across algorithms, theory, and applications, this review aims to clarify both the opportunities and the limitations of quantum machine learning, and to help guide future efforts toward robust and impactful quantum-enabled learning systems.

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Author contributions

Conceptualization, V.S., A.S.B., M.K.S., M.S. and S.K.; writing—original draft preparation, V.S., A.S.B., M.K.S. and M.S.; writing—review and editing, V.S., A.S.B., M.K.S. and M.S.; supervision, S.K.; funding acquisition, S.K. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare that they have no competing interests.

Data availability statement

No new data were created or analyzed in this study. Data sharing is not applicable to this article.

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